MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

*OSCILLATOR, DIGITAL, KEY CODE HVA*

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic oscillator module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table III.

| | | |
|---|---|---|
| | Name | Oscillator |
| | Family | Digital |
| | Key code | HVA |
| | Size | 1A |
| | Weight | 35 grams maximum |
| | Failure rate | 0.22 failures/$10^6$ hours maximum |
| * | Environment | Class I as specified in MIL-STD-1389 |
| | *Supply voltage* | 5 Vdc ±10 percent |
| | Maximum power dissipation | 1,150 mW (outputs not loaded)<br>1,500 mW (outputs loaded) |
| | Input voltage | |
| | Low level | -0.5 V minimum, 0.8 V maximum |
| | High level | 2.0 V minimum, $V_{CC}$ maximum |
| | Input current | |
| | G1-G4 | |
| | Low level ($V_{IL}$ = 0.0 V) | -4.4 mA maximum |
| | ($V_{IL}$ = 0.5 V) | -4.0 mA maximum |
| | *High level* ($V_{IH}$ = 2.5 V) | *100 µA maximum* |
| | ($V_{IH}$ = 5.5 V) | 1.0 mA maximum |
| | FPC | |
| | Low level ($V_{IL}$ = 0.0 V) | -21.5 mA maximum |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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| | |
|---|---|
| Output voltage | |
| WE, $\overline{WE}$ | |
| Low level ($I_{OL}$ = 20 mA) | 0.5 V maximum |
| High level ($I_{OH}$ = -1.0 mA) | 2.5 V minimum |
| OUT 1A, 1B-4A, 4B | |
| Low level ($I_{OL}$ = 60 mA) | 0.5 V maximum |
| High level ($I_{OH}$ = -3.0 mA) | 2.5 V minimum |
| ($I_{OH}$ = -40 mA) | 2.0 V minimum |
| Output drive current maximum capability | |
| WE, $\overline{WE}$ | |
| Low level ($V_{OL}$ = 0.5 V) | 20 mA |
| High level ($V_{OH}$ = 2.5 V) | -1.0 mA |
| OUT 1A, 1B-4A, 4B | |
| Low level ($V_{OL}$ = 0.5 V) | 60 mA |
| High level ($V_{OH}$ = 2.5 V) | -3.0 mA |
| ($V_{OH}$ = 2.0 V) | -40 mA |
| Frequency | |
| FPC (pin 16) open | |
| OUT 1A, 1B-3A, 3B, WE, $\overline{WE}$ | 3.8333 MHz ±0.01 percent |
| OUT 4A, 4B | 7.6666 MHz ±0.01 percent |
| FPC (pin 16) at 0 V | |
| OUT 1A, 1B-4A, 4B, WE, $\overline{WE}$ | 11.5000 MHz ±0.01 percent |
| Pulse width | |
| FPC (pin 16) open | |
| Low level | |
| OUT 1A, 1B-4A, 4B | 40.0 ns minimum, 50.0 ns maximum |
| $\overline{WE}$ | 79.0 ns minimum, 95.0 ns maximum |
| High level | |
| WE | 79.0 ns minimum, 95.0 ns maximum |
| FPC (pin 16) at 0 V | |
| OUT 1A, 1B-4A, 4B, WE, $\overline{WE}$ | 40.0 ns minimum, 50.0 ns maximum |
| Output to output delay time | |
| FPC (pin 16) open | |
| Between OUT 1A, OUT 2A, OUT 3A | -2.0 ns minimum, 2.0 ns maximum |
| From OUT 1A, OUT 2A, OUT 3A to OUT 4A | -2.0 ns minimum, 2.0 ns maximum |
| * From OUT 1A, OUT 2A, OUT 3A to WE | -14.0 ns minimum, 2.0 ns maximum |
| * From WE to $\overline{WE}$ | |
| WE (low to high) to $\overline{WE}$ (high to low) | -4.0 ns minimum, 4.0 ns maximum |
| WE (high to low) to $\overline{WE}$ (low to high) | -4.0 ns minimum, 2.5 ns maximum |
| FPC (pin 16) at 0 V | |
| Between OUT 1A, OUT 2A, OUT 3A, OUT 4A | -3.0 ns minimum, 3.0 ns maximum |
| Propagation delay time ($t_{PLH}$) | |
| Gate to output ($C_L$ = 50 pF) | 2.0 ns minimum, 9.5 ns maximum |
| ($C_L$ = 150 pF) | 4.0 ns minimum, 13.5 ns maximum |



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* 1.3 Module function. This module contains a crystal oscillator with four synchronous line driver outputs for driving coaxial line. The module has two frequency modes controlled by the Frequency Programming Control (FPC). A gating input (G1-G4) for each synchronous line driver output is provided. WE and $\overline{WE}$ outputs are available along with the synchronous outputs.

1.3.1 Input to output relationship. Each oscillator has the input to output relationship of tables I and II for positive logic. The time relationship of the outputs is shown on figure 1.

TABLE I. Frequency control input to output relationship.

| Inputs | | | | | Outputs (MHz) | | | | |
|---|---|---|---|---|---|---|---|---|---|
| FPC | G1 | G2 | G3 | G4 | 1A,1B | 2A,2B | 3A,3B | 4A,4B | WE,$\overline{WE}$ |
| Open | H | H | H | H | 3.8333 | 3.8333 | 3.8333 | 7.6666 | 3.8333 |
| GND | H | H | H | H | 11.5000 | 11.5000 | 11.5000 | 11.5000 | 11.5000 |

NOTES:
1. H = High level.
2. L = Low level.
3. GND = 0 V REF.

TABLE II. Gate input to output relationship.

| Gate input | Output |
|---|---|
| H | Output per table I |
| L | H |

NOTES:
1. H = High level.
2. L = Low level.

1.3.2 Frequency programming control. The frequency programming control (FPC) is not a logic input and should not be driven by any logic circuit. The only connection to the FPC input shall be a ground (0 volts) to select the frequency outputs per table I.

1.3.3 Parallel outputs. Paralleling of outputs is not permissible.

1.3.4 Ground pins. All electrical parameters are measured with pins 3, 6, 7, 10, 12, 15, 17, 23, 24, 25, 26, 27, 28, 29, 30, 32, 33, 34, 35, 37, 38, and 39 connected separately to 0 volt return. System problems may result from a different grounding method. Pins 20 and 40 are also connected to the listed ground pins and may be connected to the 0 volt return at the option of the user.



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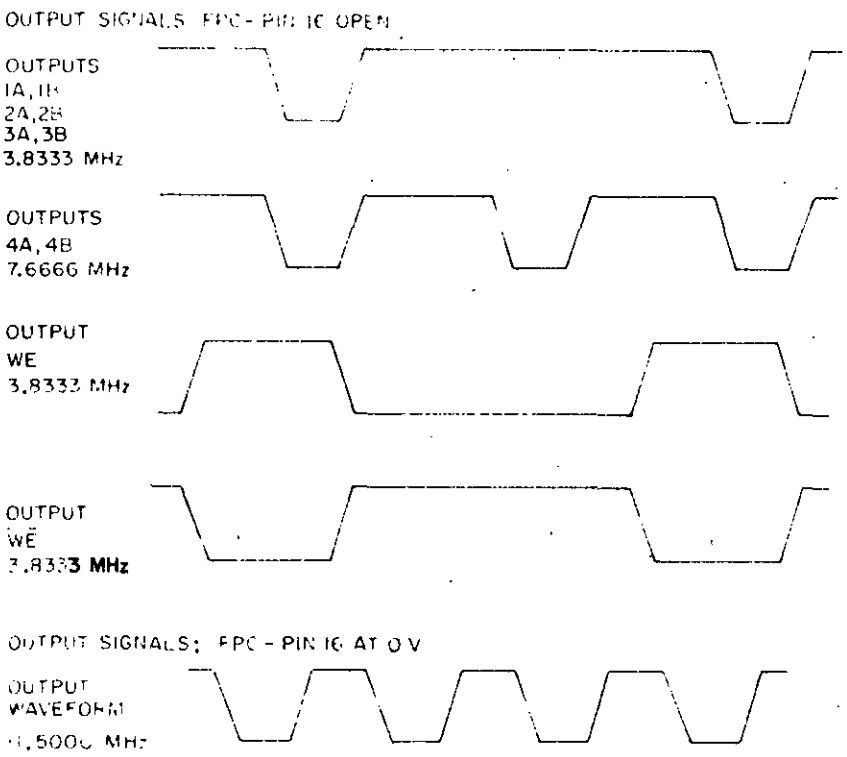



FIGURE 1. Output signal relationship.



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2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARD

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

* 3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.22 failures per million operating hours when predicted in accordance with MIL-STD-1389.

* 3.4 Performance requirements. The module shall conform to the electrical requirements specified in table III and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be HVA.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.



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3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/147-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 2.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table III.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table III (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table III for a power supply voltage ($V_{CC}$) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

| Equipment | MIL-STD-1665 designator | No. required (maximum per test figure) |
|---|---|---|
| DC power supply | Item P01 | 3 |
| Megohm bridge | Item R06 | 1 |
| Counter-timer | Item T01 | 1 |
| DC ammeter | Item X01 | 1 |
| DC voltmeter | Item V03 | 1 |
| Resistance bridge | Item R02 | 1 |
| Oscilloscope | Item D05 | 1 |
| Oscilloscope probe | Item Y01 | 2 |
| Pulse generator | Item J05 | 1 |
| Capacitance meter | Item C05 | 1 |
| Capacitance bridge | Item C01 | 1 |
| Resistors | | |
| 1 kΩ ±5%, 1/2 W | | 1 |
| 249Ω ±1%, 1/4 W | | 2 |
| 590Ω ±1%, 1/4 W | | 1 |
| 51Ω ±1%, 1/2 W | | 1 |



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| Resistors -Continued | No. required (maximum per test figure) |
|---|---|
| Value as required for figure 13, note 1b, 1/4 W | 2 |
| 2.49 kΩ ±1%, 1/8 W | 2 |
| Value as required for figure 13, note 2b, 1/4 W | 4 |
| 49.9Ω ±1%, 1/4 W | 4 |
| Value as required for figure 13, note 3b, 1/4 W | 4 |
| 825Ω ±1%, 1/4 W | 4 |
| 51.1Ω ±1%, 1/4 W | 1 |
| **Capacitors** | |
| Value as required for figure 13, note 1d, test type S | 2 |
| Value as required for figure 13, note 1d, test type Q | 2 |
| Value as required for figure 13, note 2d, test type S | 4 |
| Value as required for figure 13, note 2d, test type Q | 4 |
| **Diode** | |
| 1N3600, 1N4150, or equivalent | 6 |

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table III.

* 4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through GG shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

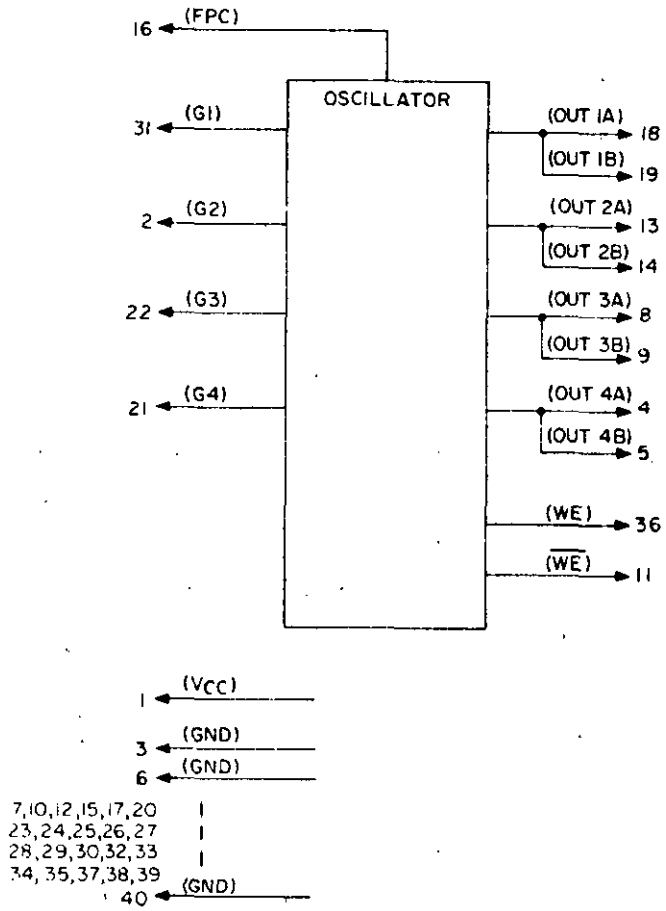
4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through GG in table III shall be performed in the order specified in 4.4.2 using the initial limits at 25° +5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through FF shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table III which are marked with an "X" in the test type column, and test condition FF, shall be performed using the initial limits at 25° ±5°C ambient.



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NOTES:
1. FPC = Frequency programming control.
2. G1-G4 = Gating inputs.
3. OUT 1A, 1B, 2A, 2B, 3A, 3B, 4A, 4B = Synchronous outputs.
4. WE, $\overline{WE}$ = Outputs.
5. $V_{CC}$ = 5.0 Vdc ±10%.
6. GND = 0 V REF.

FIGURE 2. HVA module function diagram, oscillator.



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TABLE III. Module electrical requirements.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | Acceptance requirements 25° ±5°C ambient temperature | | | | Operating temperature 0° to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| A | Isolation | IV | S | 10 | ∞ | 1.0 | ∞ | 10 | ∞ | 1.0 | ∞ | MΩ |
| B1-B5 | $V_{CC}$ stress (output frequency) | V | S | 3.7833 | 3.8833 | 3.7833 | 3.8833 | 3.7833 | 3.8833 | 3.7833 | 3.8833 | MHz |
| B6 | $V_{CC}$ stress (output frequency) | V | S | 7.5666 | 7.7666 | 7.5666 | 7.7666 | 7.5666 | 7.7666 | 7.5666 | 7.7666 | MHz |
| B7-B12 | $V_{CC}$ stress (output frequency) | V | S | 11.3500 | 11.6500 | 11.3500 | 11.6500 | 11.3500 | 11.6500 | 11.3500 | 11.6500 | MHz |
| C1-C4 | Input breakdown current, G1-G4 | VI | S | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | mA |
| C5 | Input breakdown current, FPC | VI | S | 0.0 | 2.0 | 0.0 | 2.0 | 0.0 | 2.0 | 0.0 | 2.0 | mA |
| D1-D4 | Output leakage current | VI | X | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | mA |
| E1-E4 | Output short circuit current, OUT 1A, 1B, 2A, 2B, 3A, 3B, 4A, 4B | VII | X | -50 | -225 | -50 | -225 | -50 | -225 | -50 | -225 | mA |
| E5, E6 | Output short circuit current check, WE, $\overline{WE}$ (measuring output frequency) | VII | X | 11.3500 | 11.6500 | 11.3500 | 11.6500 | 11.3500 | 11.6500 | 11.3500 | 11.6500 | MHz |
| F1-F4 | Input clamp diode voltage | VIII | X | -0.4 | -1.2 | -0.4 | -1.2 | -0.4 | -1.2 | -0.4 | -1.2 | V |
| G1-G23 | Ground continuity | IX | X | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | Ω |
| G24-G46 | Ground continuity | IX | Q | 0.0 | 0.10 | 0.0 | 0.10 | 0.0 | 0.10 | 0.0 | 0.10 | Ω |
| H1-H6 | Output voltage, low level | X | X | 0.0 | 0.5 | 0.0 | 0.5 | 0.0 | 0.5 | 0.0 | 0.5 | V |
| J1-J6 | Output voltage, high level, OUT 1A, 2A, 3A, 4A, WE, $\overline{WE}$ | XI | X | 2.5 | 4.5 | 2.5 | 4.5 | 2.5 | 4.5 | 2.5 | 4.5 | V |
| J7-J10 | Output voltage, high level, OUT 1B, 2B, 3B, 4B | XI | X | 2.0 | 4.5 | 2.0 | 4.5 | 2.0 | 4.5 | 2.0 | 4.5 | V |

See footnotes at end of table.





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TABLE III. Module electrical requirements - Continued.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | Acceptance requirements 25° ±5°C ambient temperature | | | | Operating temperature 0° to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| K1-K4 | Input current, high level, G1-G4 | XII | X | 0 | 100 | 0 | 100 | 0 | 100 | 0 | 100 | µA |
| K5 | Input current, high level, FPC | XII | X | -2.5 | -3.3 | -2.2 | -4.0 | -2.2 | -4.0 | -2.2 | -4.0 | mA |
| L1-L4 | Input current, low level, G1-G4 | XIII | X | -0.1 | -4.4 | -0.1 | -4.4 | -0.1 | -4.4 | -0.1 | -4.4 | mA |
| L5 | Input current, low level, FPC | XIII | X | -5.4 | -21.2 | -5.4 | -21.2 | -5.2 | -21.5 | -5.2 | -21.5 | mA |
| M | Power supply current | XIV | X | 50 | 209 | 50 | 209 | 50 | 209 | 50 | 209 | mA |
| N1-N5 | Frequency | XV | X | 3.83292 | 3.83368 | 3.83292 | 3.83368 | 3.83292 | 3.83368 | 3.83292 | 3.83368 | MHz |
| N6 | Frequency | XV | X | 7.66583 | 7.66737 | 7.66583 | 7.66737 | 7.66583 | 7.66737 | 7.66583 | 7.66737 | MHz |
| N7-N12 | Frequency | XV | X | 11.49885 | 11.50115 | 11.49885 | 11.50115 | 11.49885 | 11.50115 | 11.49885 | 11.50115 | MHz |
| P1-P6 | Transition time ($t_{THL}$), $C_L$ = 50 pF, OUT 1B, 2B, 3B, 4B, WE, $\overline{WE}$ | XVI | S | 2.0 | 6.0 | 2.0 | 6.0 | 2.0 | 8.0 | 2.0 | 8.0 | ns |
| P7-P12 | Transition time ($t_{THL}$), $C_L$ = 150 pF, OUT 1B, 2B, 3B, 4B, WE, $\overline{WE}$ | XVI | Q | 2.0 | 11.0 | 2.0 | 11.0 | 2.0 | 13.0 | 2.0 | 13.0 | ns |
| Q1-Q6 | Transition time ($t_{TLH}$), $C_L$ = 50 pF, OUT 1A, 2A, 3A, 4A, WE, $\overline{WE}$ | XVI | S | 2.0 | 15.0 | 2.0 | 15.0 | 2.0 | 18.0 | 2.0 | 18.0 | ns |
| Q7-Q12 | Transition time ($t_{TLH}$), $C_L$ = 150 pF, OUT 1A, 2A, 3A, 4A, WE, $\overline{WE}$ | XVI | Q | 2.0 | 27.0 | 2.0 | 27.0 | 2.0 | 30.0 | 2.0 | 30.0 | ns |
| R1-R4 | Pulse width ($t_{PW}$), OUT 1A, 2A, 3A, 4A | XVI | X | 40.0 | 49.5 | 40.0 | 49.5 | 40.0 | 50.0 | 40.0 | 50.0 | ns |
| R5, R6 | Pulse width ($t_{PW}$), WE, $\overline{WE}$ | XVI | X | 80.0 | 94.0 | 80.0 | 94.0 | 79.0 | 95.0 | 79.0 | 95.0 | ns |

See footnotes at end of table.



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TABLE III. Module electrical requirements - Continued.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | Acceptance requirements 25° ±5°C ambient temperature | | | | Operating temperautre 0° to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| R7-R12 | Pulse width ($t_{PW}$), OUT 1B, 2B, 3B, 4B, WE, $\overline{WE}$ | XVI | X | 40.0 | 49.5 | 40.0 | 49.5 | 40.0 | 50.0 | 40.0 | 50.0 | ns |
| S1-S3 | Output to output delay time ($t_{DELAY1-LH}$) | XVII | X | -2.0 | 2.0 | -2.0 | 2.0 | -2.0 | 2.0 | -2.0 | 2.0 | ns |
| T1-T3 | Output to output delay time ($t_{DELAY1-HL}$) | XVII | X | -2.0 | 2.0 | -2.0 | 2.0 | -2.0 | 2.0 | -2.0 | 2.0 | ns |
| U1-U3 | Output to output delay time ($t_{DELAY2-LH}$) | XVII | X | -2.0 | 2.0 | -2.0 | 2.0 | -2.0 | 2.0 | -2.0 | 2.0 | ns |
| V1-V3 | Output to output delay time ($t_{DELAY2-HL}$) | XVII | X | -2.0 | 2.0 | -2.0 | 2.0 | -2.0 | 2.0 | -2.0 | 2.0 | ns |
| * W1-W3 | Output to output delay time ($t_{DELAY3}$) | XVII | X | -14.0 | 2.0 | -14.0 | 2.0 | -14.0 | 4.0 | -14.0 | 4.0 | ns |
| * X | Output to output delay time ($t_{DELAY4}$) | XVII | X | -4.0 | 4.0 | -4.0 | 4.0 | -4.0 | 4.0 | -4.0 | 4.0 | ns |
| Y | Output to output delay time ($t_{DELAY5}$) | XVII | X | -3.5 | 2.5 | -3.5 | 2.5 | -4.0 | 2.5 | -4.0 | 2.5 | ns |
| Z1-Z6 | Output to output delay time ($t_{DELAY6-LH}$) | XVII | X | -3.0 | 3.0 | -3.0 | 3.0 | -3.0 | 3.0 | -3.0 | 3.0 | ns |
| AA1-AA6 | Output to output delay time ($t_{DELAY6-HL}$) | XVII | X | -3.0 | 3.0 | -3.0 | 3.0 | -3.0 | 3.0 | -3.0 | 3.0 | ns |
| BB1-BB4 | Propagation delay time ($t_{PLH}$), gate to output, $C_L$ = 50 pF | XVIII | S | 2.0 | 8.5 | 2.0 | 8.5 | 2.0 | 9.5 | 2.0 | 9.5 | ns |

See footnotes at end of table.





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TABLE III. Module electrical requirements - Continued.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | Acceptance requirements 25° ±5°C ambient temperature | | | | Operating temperature 0° to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| BB5-BB8 | Propagation delay time ($t_{PLH}$), gate to output, $C_L$ = 150 pF | XVIII | Q | 4.0 | 12.5 | 4.0 | 12.5 | 4.0 | 13.5 | 4.0 | 13.5 | ns |
| CC1-CC6 | Output crosstalk, high level | XIX | Q | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | mV |
| DD1-DD6 | Output crosstalk, low level | XIX | Q | 0 | 1.0 | 0 | 1.0 | 0 | 1.0 | 0 | 1.0 | V |
| EE | Power supply transient amplitude | XX | Q | 0 | 500 | 0 | 500 | 0 | 500 | 0 | 500 | mV |
| FF | Power supply filter capacitance | XXI | S | 4.4 | 11.6 | 4.4 | 11.6 | 4.4 | 11.6 | 4.4 | 11.6 | µF |
| GG1-GG4 | Input capacitance | XXII | Q | 5.0 | 20.0 | 5.0 | 20.0 | 5.0 | 20.0 | 5.0 | 20.0 | pF |

NOTES:
1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.



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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test, electrical tests D through FF shall be performed in the order specified in 4.4.2 using the 0° to 60°C operating temperature end-of-life requirements of table III. After the module has been returned to 25°C, electrical tests D through GG shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table III.

5. PACKAGING

* 5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table III, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/147-1 supersedes part number M28787/147. The two parts are interchangeable.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

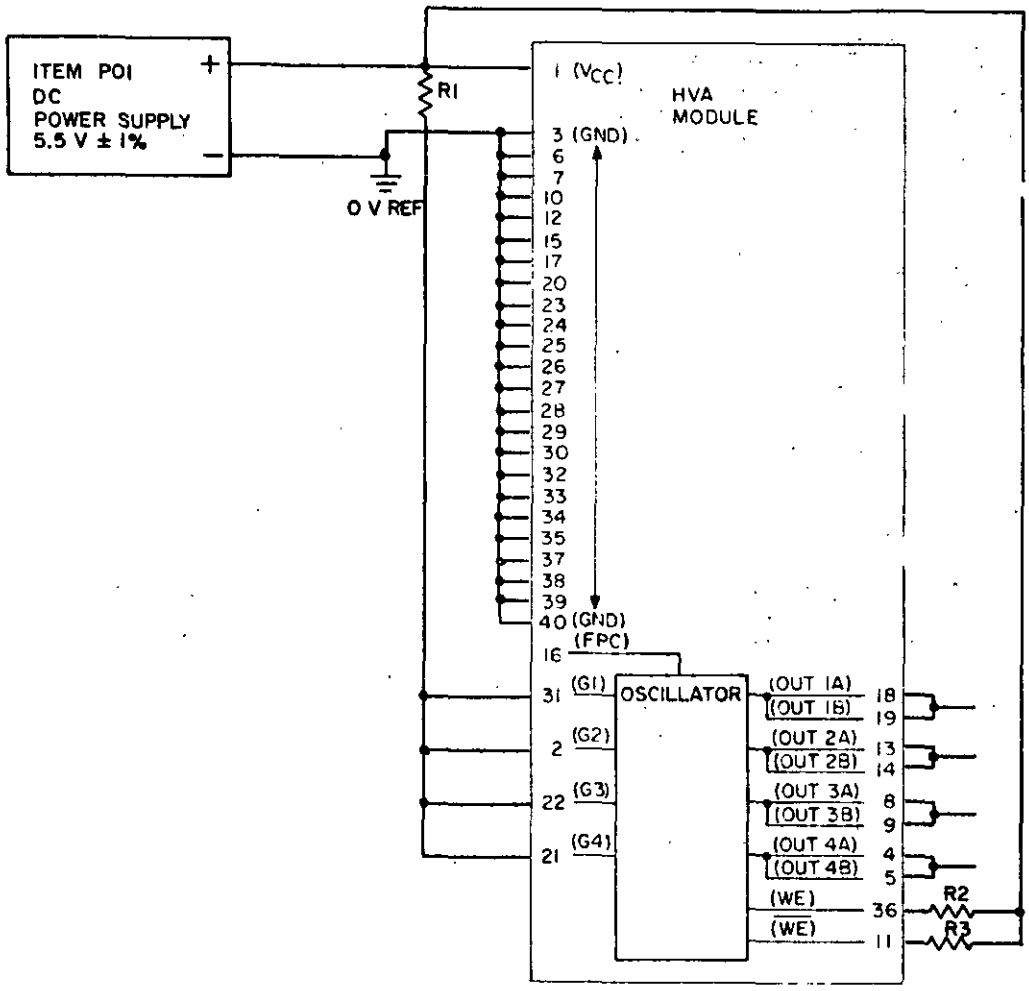
Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5963-N102)



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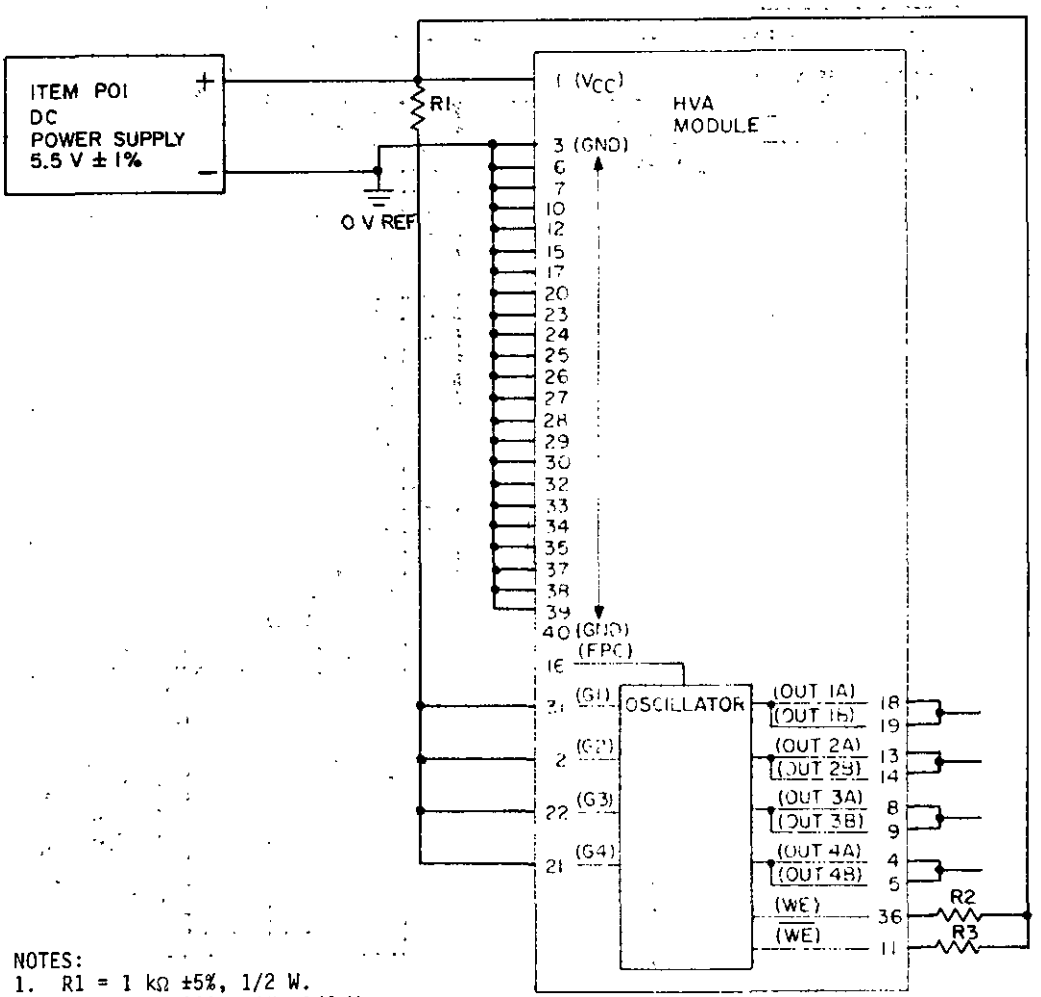



NOTES:
1. R1 = 1 kΩ ±5%, 1/2 W.
2. R2 = R3 = 249Ω ±1%, 1/4 W.
3. Connect a logic load to outputs 1,2, 3, and 4 as specified on figure 13 with the exception that capacitor $C_L$ is not required.

FIGURE 3. Operating temperature and life test circuit.



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NOTES:
1. R1 = 1 kΩ ±5%, 1/2 W.
2. R2 = R3 = 249Ω ±1%, 1/4 W.
3. Connect a logic load to outputs 1,2, 3, and 4 as specified on figure 13 with the exception that capacitor $C_L$ is not required.

FIGURE 4. Component temperature test circuit.



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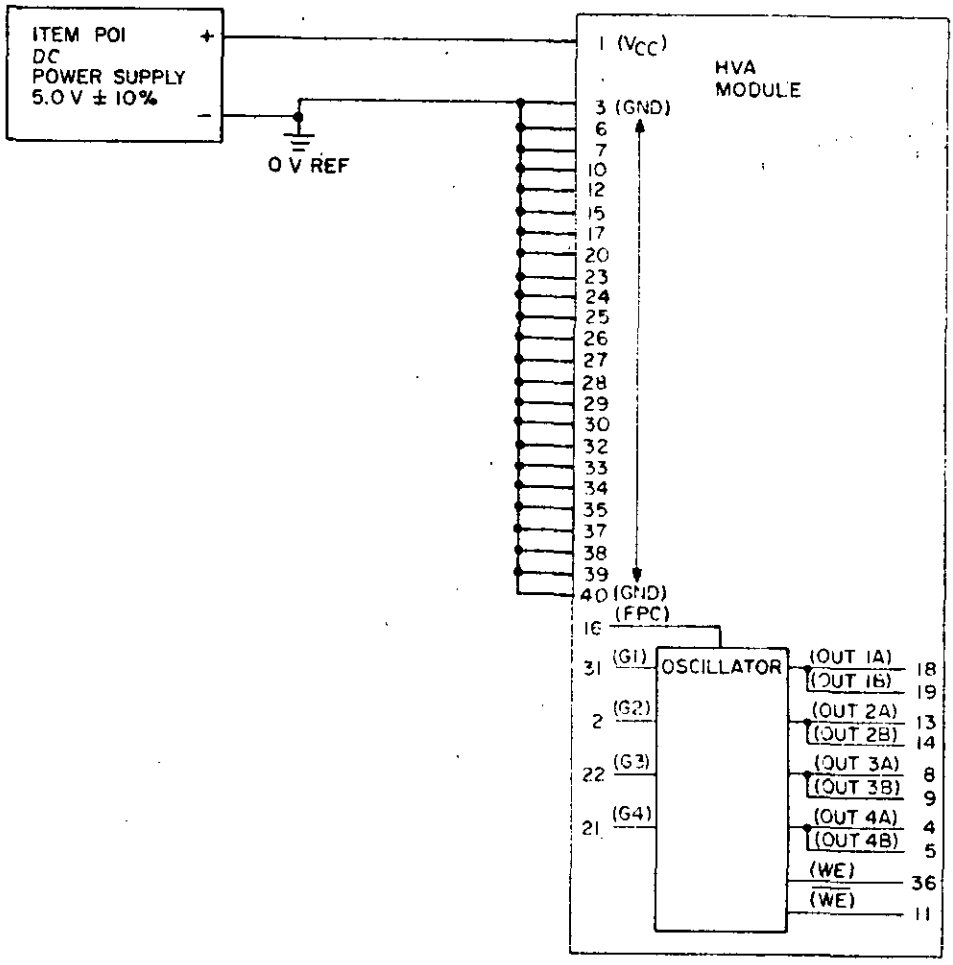



FIGURE 5. Vibration test circuit.



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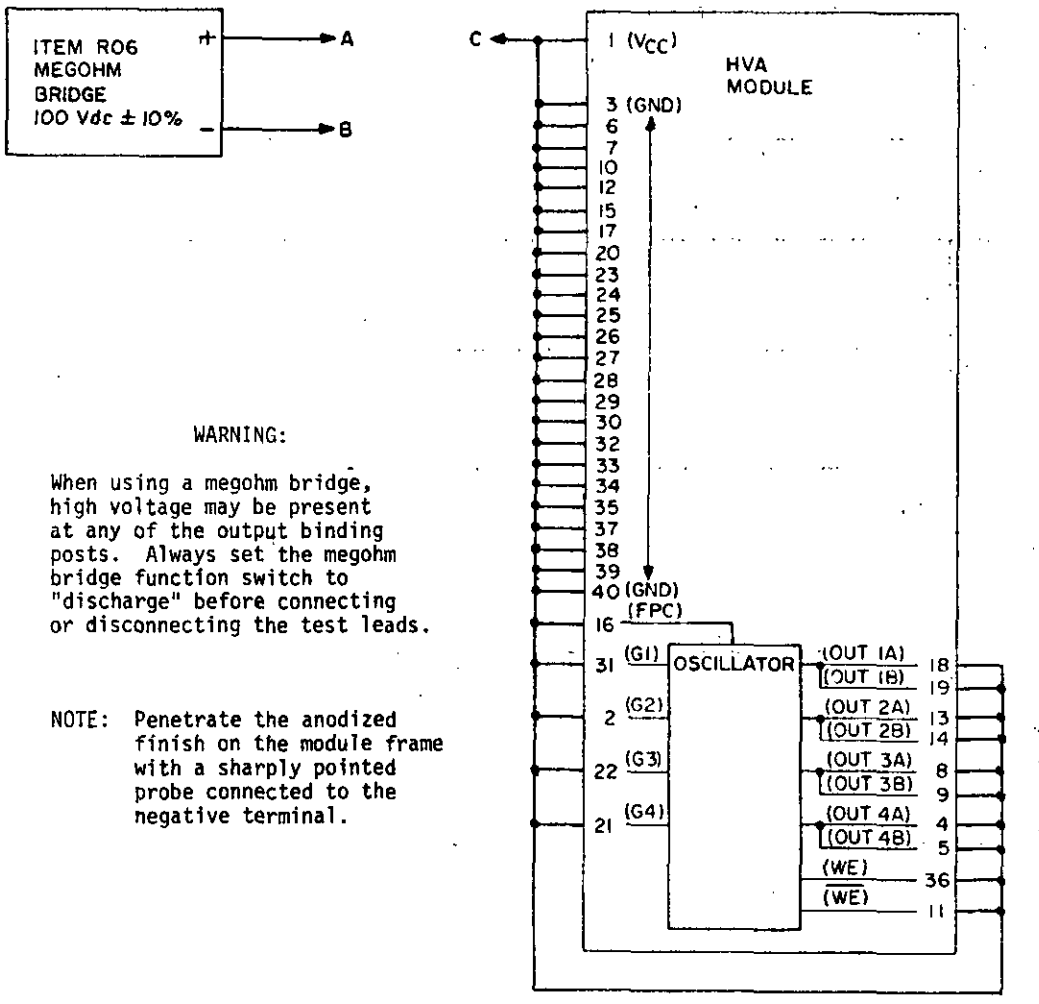
TABLE IV. Isolation test.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| A | Isolation | C | Frame | 10 | ∞ | MΩ | 6 |





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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 6. Isolation test circuit for table IV.



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TABLE V. $V_{CC}$ stress (output frequency) test.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| B1 | OUT 1A $V_{CC}$ stress (output frequency) | 18 | Open | 3.7833 | 3.8833 | MHz | 7 |
| B2 | OUT 2A | 13 | | | | | |
| B3 | OUT 3A | 8 | | | | | |
| B4 | WE | 36 | | | | | |
| B5 | $\overline{WE}$ | 11 | | 3.7833 | 3.8833 | | |
| B6 | OUT 4A | 4 | Open | 7.5666 | 7.7666 | | |
| B7 | OUT 1B | 19 | 16 | 11.3500 | 11.6500 | | |
| B8 | OUT 2B | 14 | | | | | |
| B9 | OUT 3B | 9 | | | | | |
| B10 | OUT 4B | 5 | | | | | |
| B11 | WE | 36 | | | | | |
| B12 | $\overline{WE}$ $V_{CC}$ stress (output frequency) | 11 | 16 | 11.3500 | 11.6500 | MHz | 7 |





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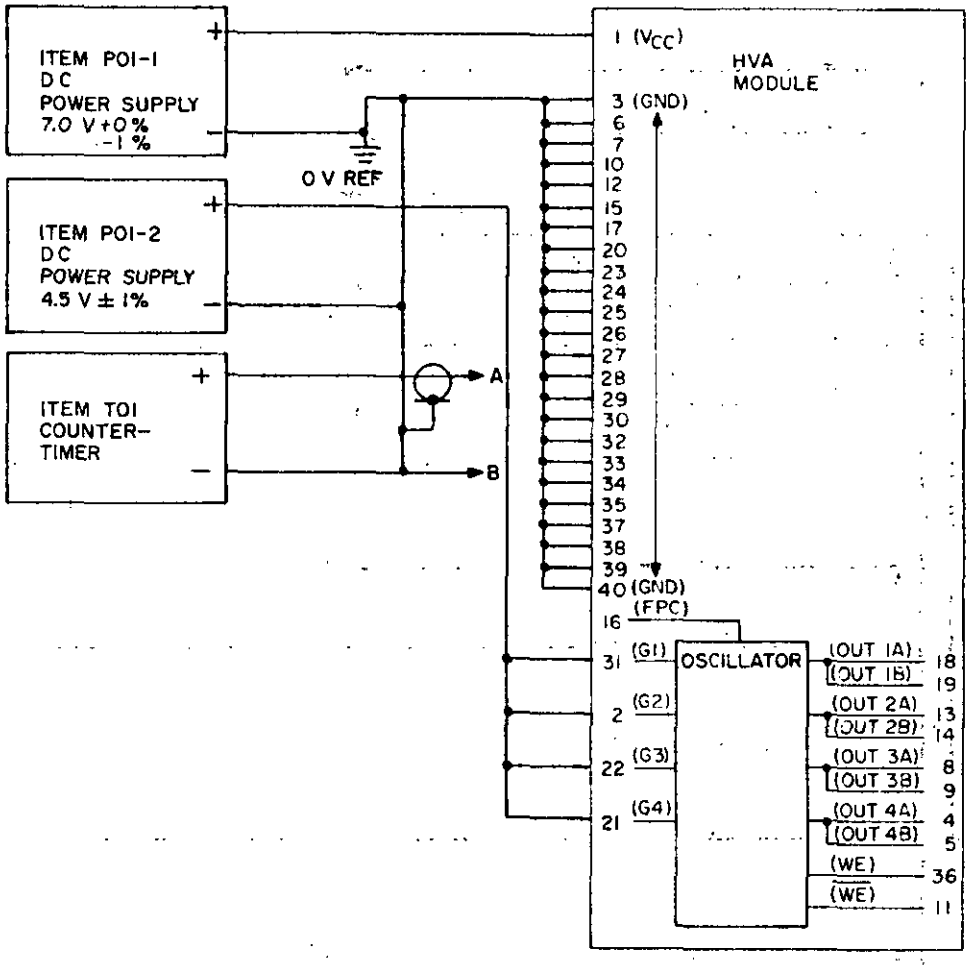



FIGURE 7. $V_{CC}$ stress (output frequency) test circuit for table V.



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TABLE VI. Input breakdown current and output leakage current test.

| Test condition | Parameter | | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | Min | Max | | |
| C1 | G1 | Input breakdown current | 2,21,22 | 31 | 0.0 | 1.0 | mA | 8 |
| C2 | G2 | | 21,22,31 | 2 | | | | |
| C3 | G3 | | 2,21,31 | 22 | | | | |
| C4 | G4 | | 2,22,31 | 21 | | 1.0 | | |
| C5 | FPC | Input breakdown current | Open | 16 | | 2.0 | | |
| D1 | OUT 1A, 1B | Output leakage current | 2,21,22,31 | 18,19 | | 1.0 | | |
| D2 | OUT 2A, 2B | | | 13,14 | | | | |
| D3 | OUT 3A, 3B | | | 8,9 | | | | |
| D4 | OUT 4A, 4B | Output leakage current | 2,21,22,31 | 4,5 | 0.0 | 1.0 | mA | 8 |





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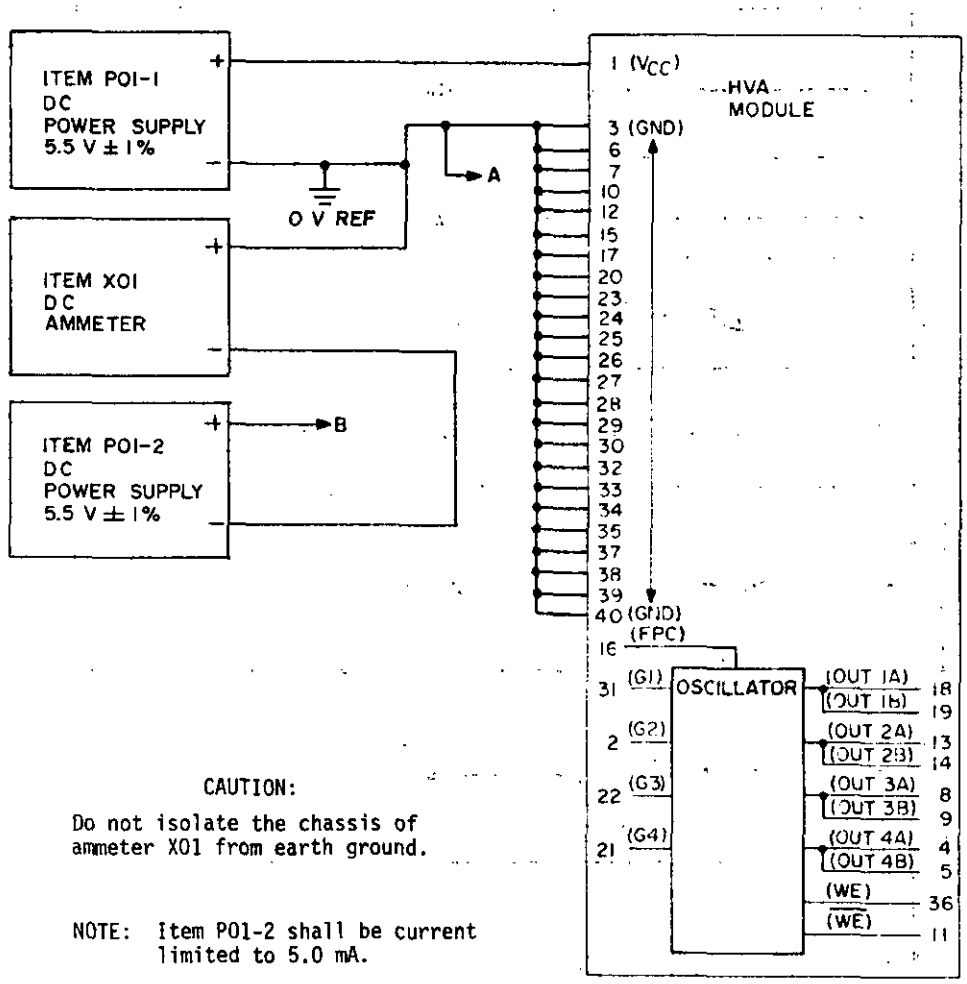



FIGURE 8. Input breakdown current and output leakage current test circuit for table VI.



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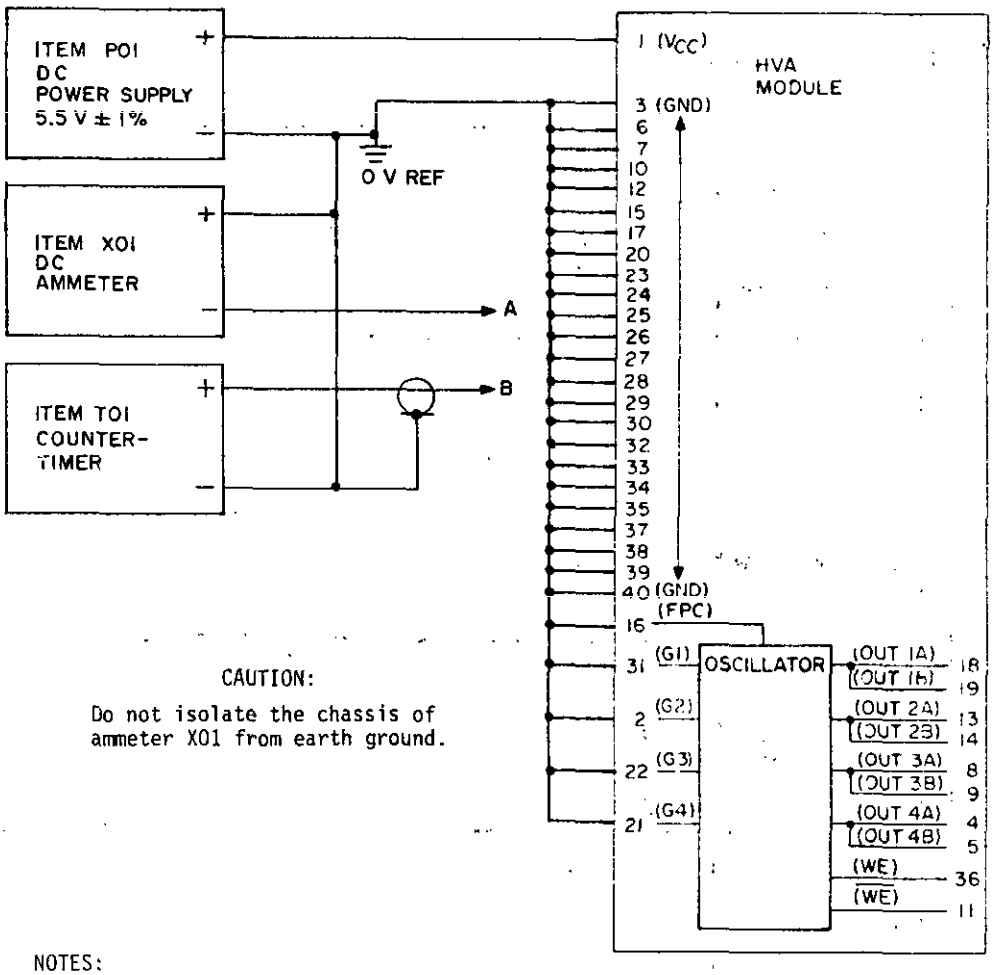
TABLE VII. Output short circuit current test and check.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| E1 | OUT 1A, 1B Output short circuit current | 18,19 | Open | -50 | -225 | mA | 9 |
| E2 | OUT 2A, 2B | 13,14 | | | | | |
| E3 | OUT 3A, 3B | 8,9 | | | | | |
| E4 | OUT 4A, 4B Output short circuit current | 4,5 | Open | -50 | -225 | mA | |
| E5 | WE Output short circuit current check | See note 2 (figure 9) | 36 | 11.3500 | 11.6500 | MHz | |
| E6 | $\overline{WE}$ Output short circuit current check | See note 2 (figure 9) | 11 | 11.3500 | 11.6500 | MHz | 9 |





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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:

1. When performing output short circuit current test, only one output shall be shorted at a time, and the duration of the test shall not exceed 1 second.
2. After the input connections for test conditions E5 and E6 have been accomplished, the output under test shall be short circuited by connecting the output to point A with the duration not to exceed 5 seconds. The short circuit shall then be removed and the output frequency measured.

FIGURE 9. Output short circuit current test and check test circuit for table VII.



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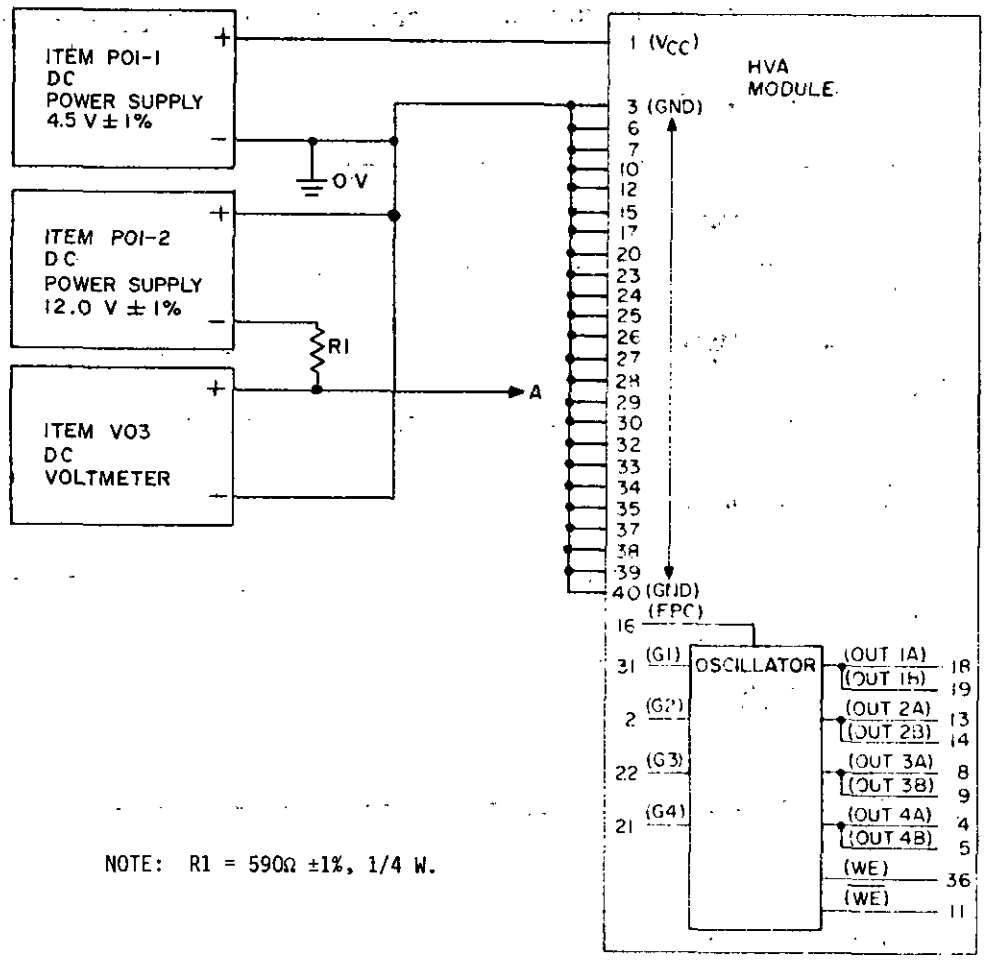
TABLE VIII. Input clamp diode voltage test.

| Test condition | Parameter | | Connect | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | | A to pin | Min | Max | | |
| F1 | G1 | Input clamp diode voltage | 31 | -0.4 | -1.2 | V | 10 |
| F2 | G2 | ↓ | 2 | ↓ | ↓ | ↓ | ↓ |
| F3 | G3 | ↓ | 22 | ↓ | ↓ | ↓ | ↓ |
| F4 | G4 | Input clamp diode voltage | 21 | -0.4 | -1.2 | V | 10 |





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NOTE: R1 = 590Ω ±1%, 1/4 W.

FIGURE 10. Input clamp diode voltage test circuit for table VIII.



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TABLE IX. Continuity test.

| Test condition | Parameter | | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | Min | Max | | |
| G1 | Pin 3 | Ground continuity | 10 | 3 | 0.0 | 1.0 | Ω | 11 |
| G2 | Pin 6 | | | 6 | | | | |
| G3 | Pin 7 | | | 7 | | | | |
| G4 | Pin 12 | | | 12 | | | | |
| G5 | Pin 15 | | | 15 | | | | |
| G6 | Pin 17 | | | 17 | | | | |
| G7 | Pin 20 | | | 20 | | | | |
| G8 | Pin 23 | | | 23 | | | | |
| G9 | Pin 24 | | | 24 | | | | |
| G10 | Pin 25 | | | 25 | | | | |
| G11 | Pin 26 | | | 26 | | | | |
| G12 | Pin 27 | | | 27 | | | | |
| G13 | Pin 28 | | | 28 | | | | |
| G14 | Pin 29 | | | 29 | | | | |
| G15 | Pin 30 | | | 30 | | | | |
| G16 | Pin 32 | | | 32 | | | | |
| G17 | Pin 33 | | | 33 | | | | |
| G18 | Pin 34 | | | 34 | | | | |
| G19 | Pin 35 | Ground continuity | 10 | 35 | 0.0 | 1.0 | Ω | 11 |





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TABLE IX. Continuity test - Continued.

| Test condition | Parameter | | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | Min | Max | | |
| G20 | Pin 37 | Ground continuity | 10 | 37 | 0.0 | 1.0 | Ω | 11 |
| G21 | Pin 38 | | | 38 | | | | |
| G22 | Pin 39 | | | 39 | | | | |
| G23 | Pin 40 | | | 40 | | 1.0 | | |
| G24 | Pin 3 | | | 3 | | 0.10 | | |
| G25 | Pin 6 | | | 6 | | | | |
| G26 | Pin 7 | | | 7 | | | | |
| G27 | Pin 12 | | | 12 | | | | |
| G28 | Pin 15 | | | 15 | | | | |
| G29 | Pin 17 | | | 17 | | | | |
| G30 | Pin 20 | | | 20 | | | | |
| G31 | Pin 23 | | | 23 | | | | |
| G32 | Pin 24 | | | 24 | | | | |
| G33 | Pin 25 | | | 25 | | | | |
| G34 | Pin 26 | | | 26 | | | | |
| G35 | Pin 27 | | | 27 | | | | |
| G36 | Pin 28 | | | 28 | | | | |
| G37 | Pin 29 | | | 29 | | | | |
| G38 | Pin 30 | Ground continuity | 10 | 30 | 0.0 | 0.10 | Ω | 11 |



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TABLE IX. Continuity test - Continued.

| Test condition | Parameter | | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | Min | Max | | |
| G39 | Pin 32 | Ground continuity | 10 | 32 | 0.0 | 0.10 | Ω | 11 |
| G40 | Pin 33 | | | 33 | | | | |
| G41 | Pin 34 | | | 34 | | | | |
| G42 | Pin 35 | | | 35 | | | | |
| G43 | Pin 37 | | | 37 | | | | |
| G44 | Pin 38 | | | 38 | | | | |
| G45 | Pin 39 | | | 39 | | | | |
| G46 | Pin 40 | Ground continuity | 10 | 40 | 0.0 | 0.10 | Ω | 11 |





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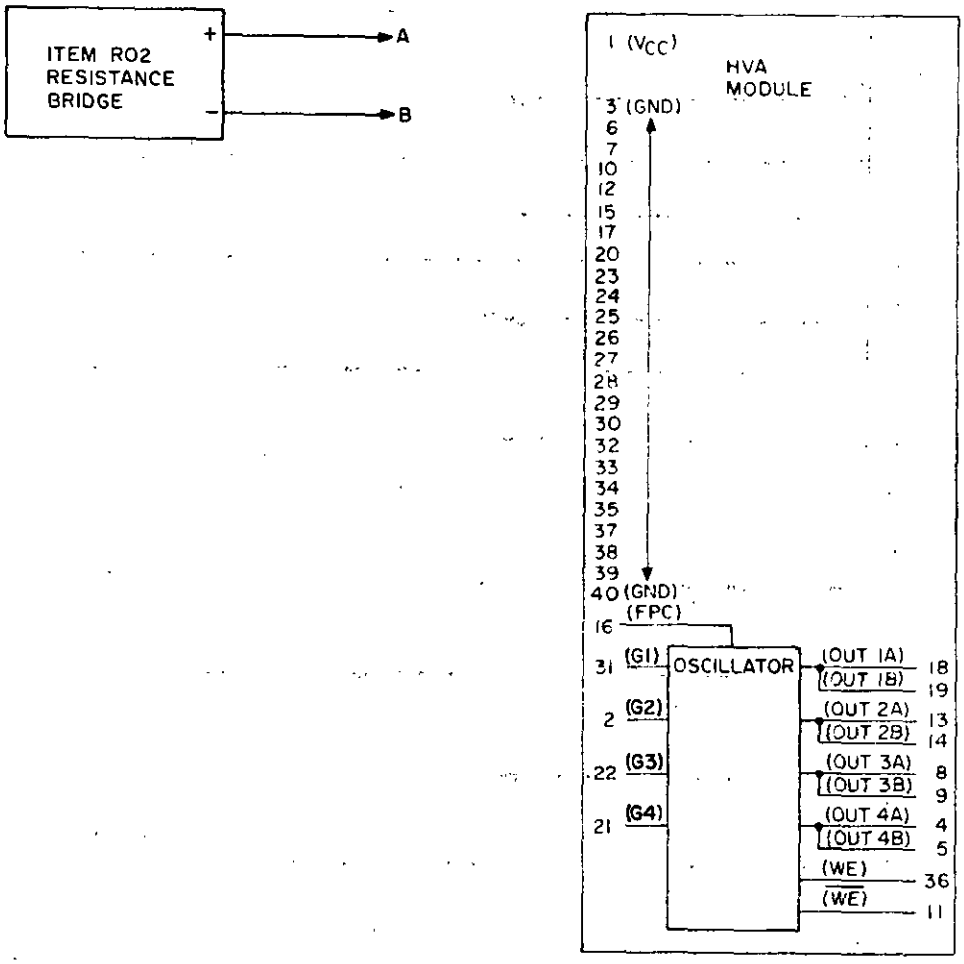



FIGURE 11. Continuity test circuit for table IX.



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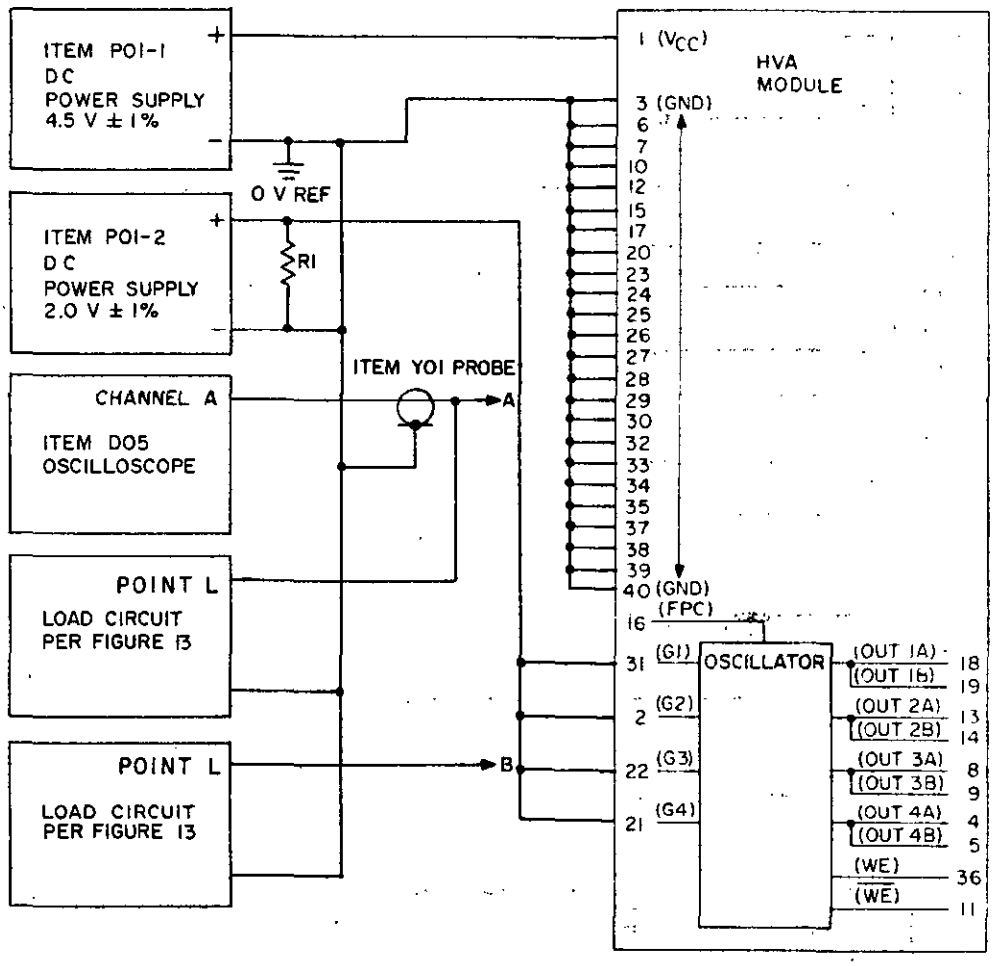
TABLE X. Output voltage (low-level) test.

| Test condition | Parameter | | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | Min | Max | | |
| H1 | OUT 1A | Output voltage, low level | 18 | Open | 0.0 | 0.5 | V | 12,14 |
| H2 | OUT 2A | | 13 | | | | | |
| H3 | OUT 3A | | 8 | | | | | |
| H4 | OUT 4A | | 4 | Open | | | | |
| H5 | WE | | 36 | 11 | | | | |
| H6 | $\overline{WE}$ | Output voltage, low level | 11 | 36 | 0.0 | 0.5 | V | 12,14 |





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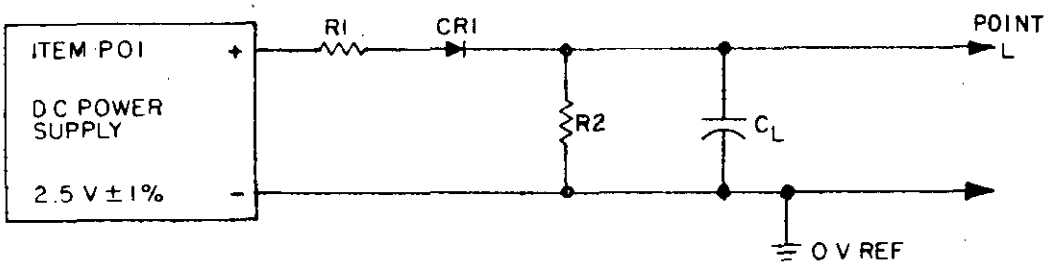



NOTE: R1 = 51Ω ±5%, 1/2 W.

FIGURE 12. Output voltage (low-level) test circuit for table X.



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NOTES:

1. Load circuit for WE and $\overline{WE}$ outputs:

   a. CR1 = 1N3600, 1N4150, or equivalent.
   b. R1 = Select value for a current flow of 20 mA ±5% with point L held at 0.5 V.
   c. R2 = 2.49 kΩ ±1%, 1/8 W.
   d. Test type S:
      $C_L$ = 50 ±5 pF (including probe and parasitic capacitance).
      Test types X and Q:
      $C_L$ = 150 ±5 pF (including probe and parasitic capacitance).
   e. Capacitor $C_L$ is not required for dc electrical test conditions H5, H6, J5, and J6.
   f. Lead length from load circuit to module output pin under test shall be less than 1 inch (25.4 mm).

2. Load circuit for OUT 1A, 1B-4A, 4B outputs for all test conditions except J1-J4 of table XI:

   a. CR1 = 1N3600, 1N4150, or equivalent.
   b. R1 = Select value for a current flow of 60 mA ±5% with point L held at 0.5 V.
   c. R2 = 49.9Ω ±1%, 1/4 W.
   d. Test type S:
      $C_L$ = 50 ±5 pF (including probe and parasitic capacitance).
      Test types X and Q:
      $C_L$ = 150 ±5 pF (including probe and parasitic capacitance).
   e. Capacitor $C_L$ is not required for dc electrical test conditions H1-H4 and J7-J10.
   f. Lead length from load circuit to module output pin under test shall be less than 1 inch (25.4 mm).
   g. Item P01 shall have a maximum of 300 mV noise at each output load point.

3. Load circuit for OUT 1A, 1B-4A, 4B outputs for test conditions J1-J4 of table XI:

   a. CR1 = 1N3600, 1N4150, or equivalent.
   b. R1 = Select value for a current flow of 60 mA ±5% with point L held at 0.5 V.
   c. R2 = 825Ω ±1%, 1/4 W.

FIGURE 13. HVA logic load circuit.



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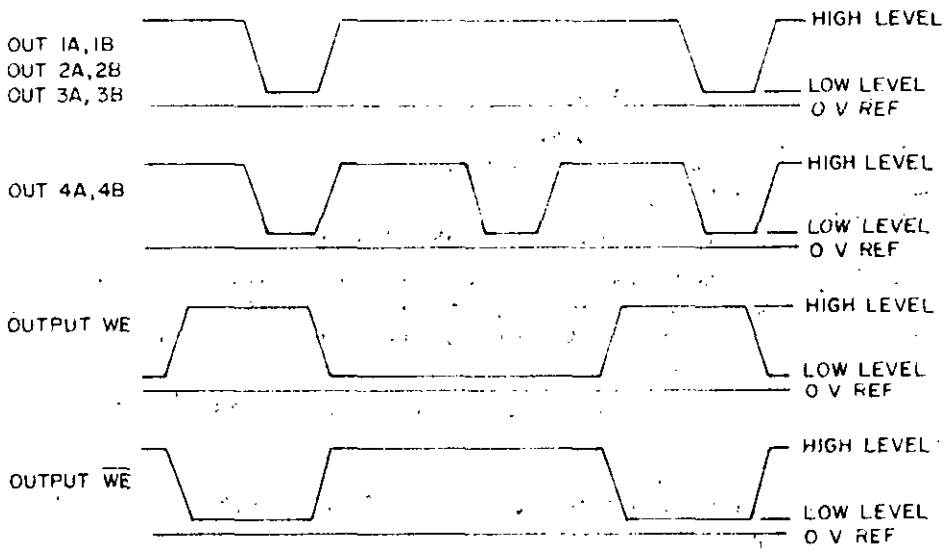



FIGURE 14. Output voltage (low-level and high-level) test circuit waveforms for tables X and XI.



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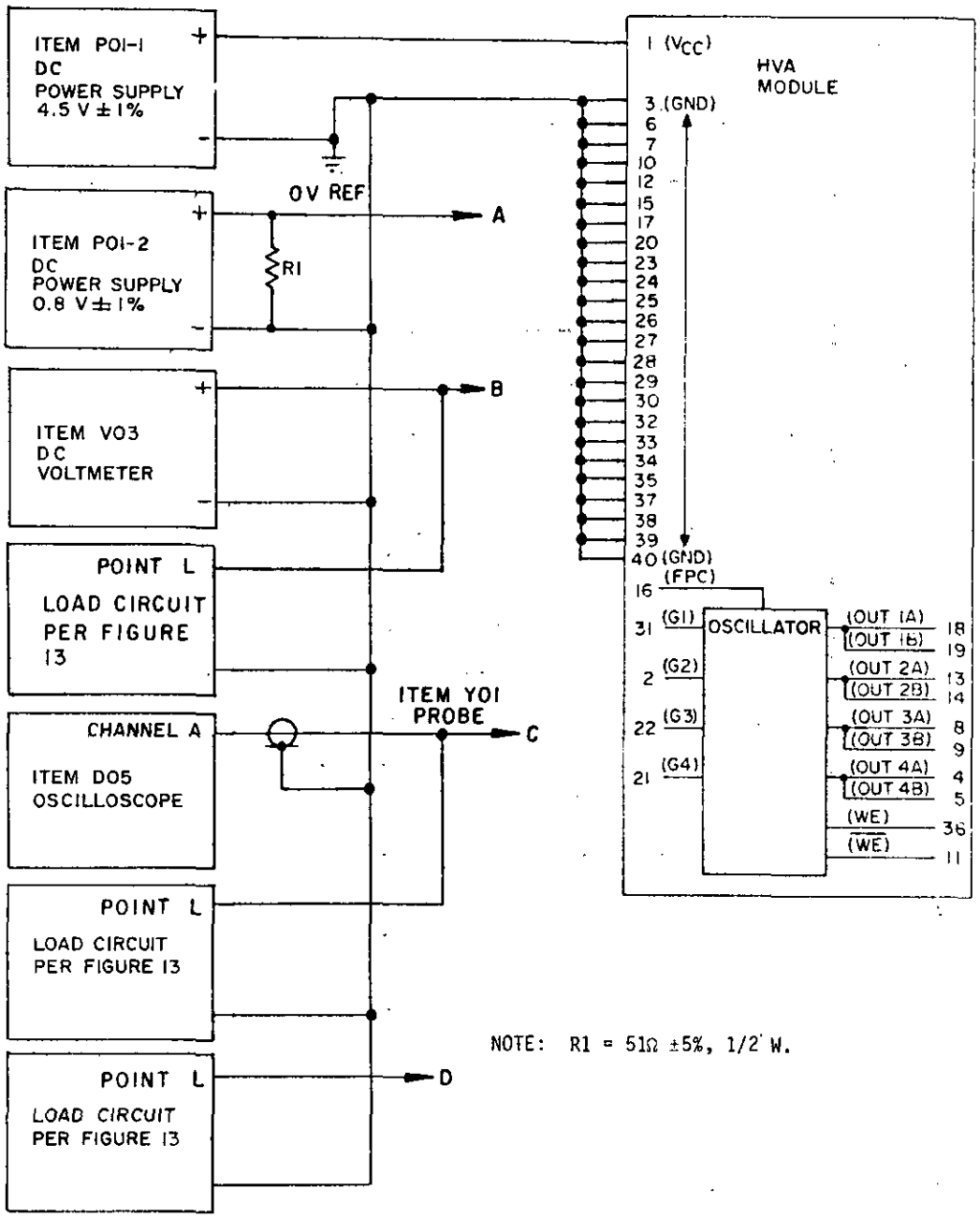
TABLE XI. Output voltage (high-level) test.

| Test condition | Parameter | | Connect | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| J1 | OUT 1A | Output voltage, high level | 31 | 18 | Open | Open | 2.5 | 4.5 | V | 14,15 |
| J2 | OUT 2A | | 2 | 13 | | | | | | |
| J3 | OUT 3A | | 22 | 8 | | | | | | |
| J4 | OUT 4A | | 21 | 4 | Open | Open | | | | |
| J5 | WE | | Open | Open | 36 | 11 | | | | |
| J6 | $\overline{WE}$ | | Open | Open | 11 | 36 | 2.5 | | | |
| J7 | OUT 1B | | 31 | 19 | Open | Open | 2.0 | | | |
| J8 | OUT 2B | | 2 | 14 | | | | | | |
| J9 | OUT 3B | | 22 | 9 | | | | | | |
| J10 | OUT 4B | Output voltage, high level | 21 | 5 | Open | Open | 2.0 | 4.5 | V | 14,15 |





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NOTE: R1 = 51Ω ±5%, 1/2 W.

* FIGURE 15. Output voltage (high-level) test circuit for table XI.



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TABLE XII. Input current (high-level) test.

| Test condition | Parameter | | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | Min | Max | | |
| K1 | G1 | Input current, high level | 31 | 2,16,21,22 | 0 | 100 | µA | 16 |
| K2 | G2 | | 2 | 16,21,22,31 | | | | |
| K3 | G3 | | 22 | 2,16,21,31 | | | | |
| K4 | G4 | | 21 | 2,16,22,31 | 0 | 100 | µA | |
| K5 | FPC | Input current, high level | 16 | Open | -2.5 | -3.3 | mA | 16 |





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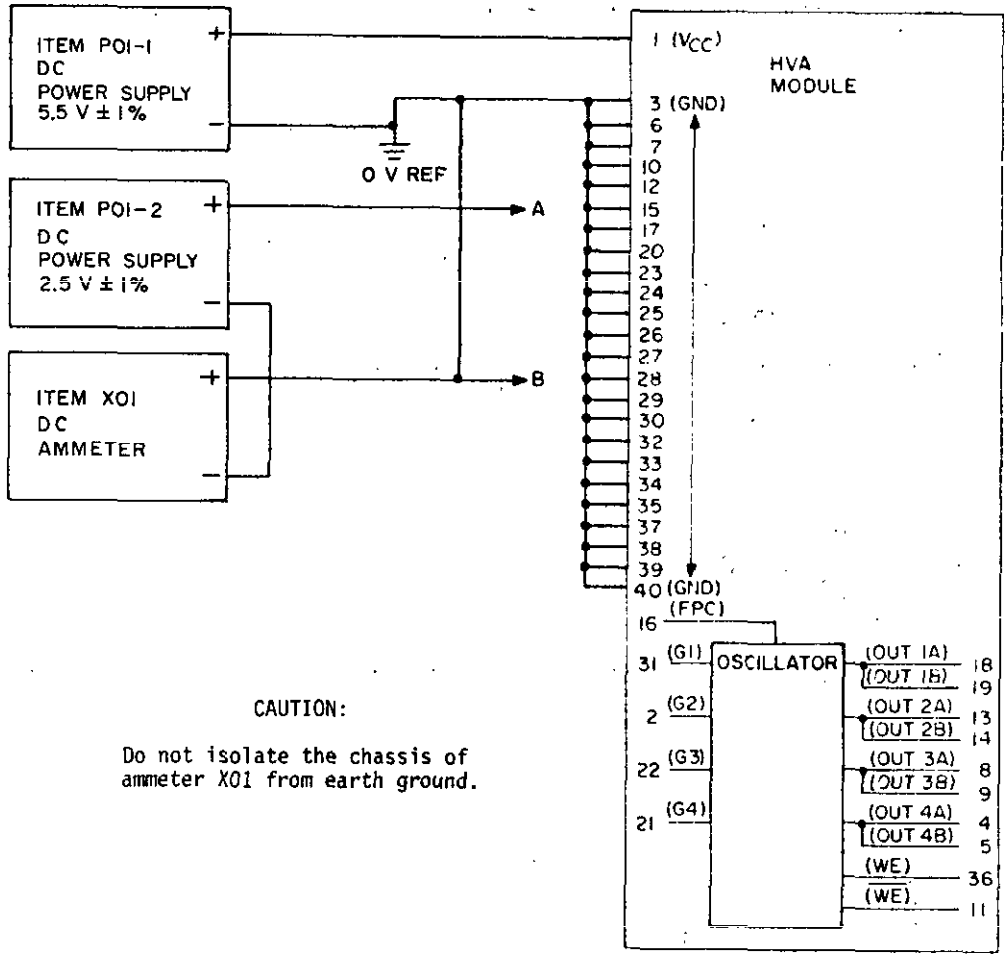



FIGURE 16. Input current (high-level) test circuit for table XII.



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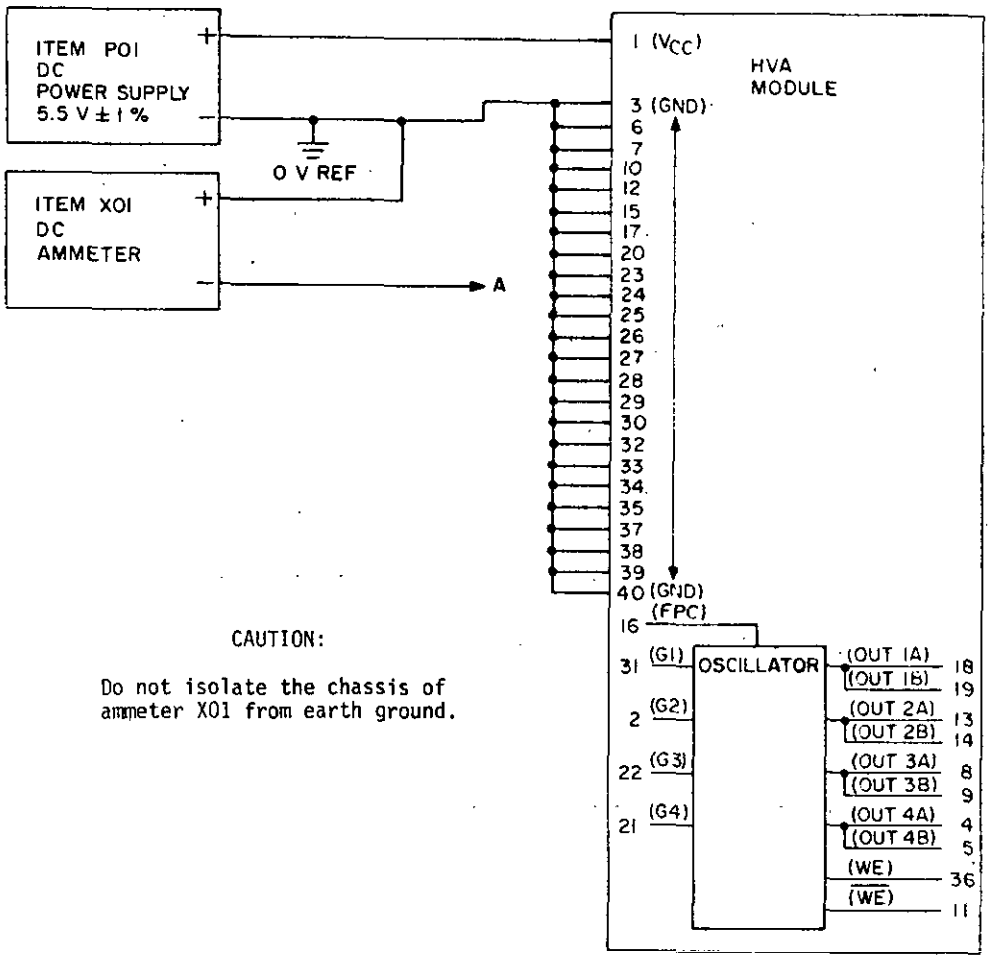
TABLE XIII. Input current (low-level) test.

| Test condition | Parameter | | Connect | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | | A to pin | Min | Max | | |
| L1 | G1 | Input current, low level | 31 | -0.1 | -4.4 | mA | 17 |
| L2 | G2 | | 2 | | | | |
| L3 | G3 | | 22 | | | | |
| L4 | G4 | | 21 | -0.1 | -4.4 | | |
| L5 | FPC | Input current, low level | 16 | -5.4 | -21.2 | mA | 17 |





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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 17. Input current (low-level) test circuit for table XIII.



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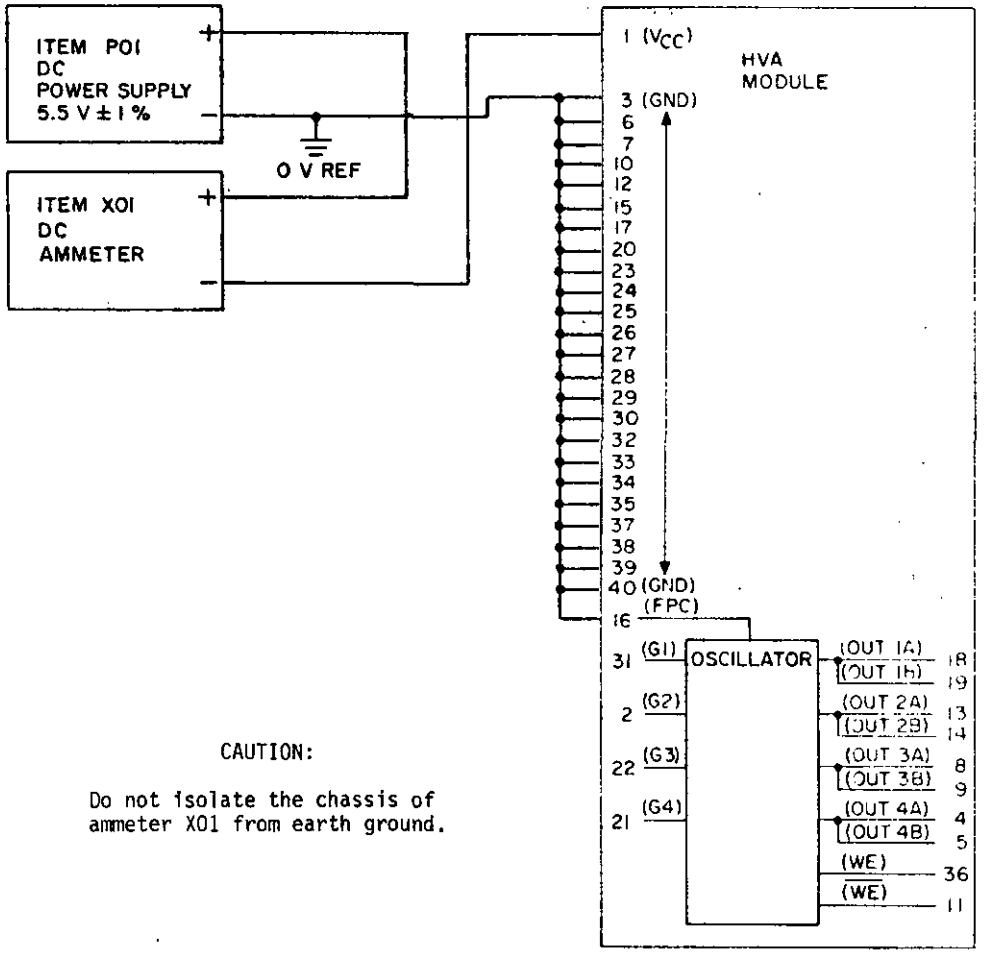
TABLE XIV. Power supply current test.

| Test condition | Parameter | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|
| | | Min | Max | | |
| M | Power supply current | 50 | 209 | mA | 18 |





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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 18. Power supply current test circuit for table XIV.



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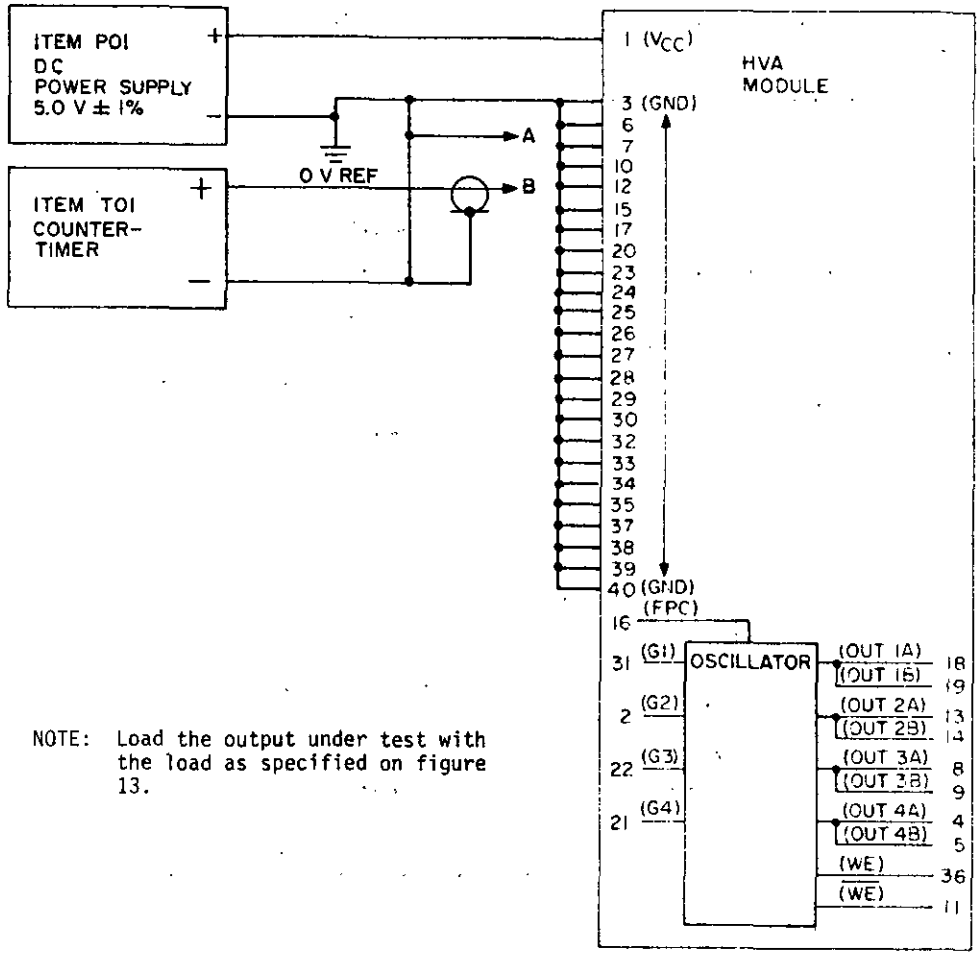
TABLE XV. Frequency test.

| Test condition | Parameter | | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | Min | Max | | |
| N1 | OUT 1A | Frequency | Open | 18 | 3.83292 | 3.83368 | MHz | 19 |
| N2 | OUT 2A | | | 13 | | | | |
| N3 | OUT 3A | | | 8 | | | | |
| N4 | WE | | | 36 | | | | |
| N5 | $\overline{WE}$ | | | 11 | 3.83292 | 3.83368 | | |
| N6 | OUT 4A | | Open | 4 | 7.66583 | 7.66737 | | |
| N7 | OUT 1B | | 16 | 19 | 11.49885 | 11.50115 | | |
| N8 | OUT 2B | | | 14 | | | | |
| N9 | OUT 3B | | | 9 | | | | |
| N10 | OUT 4B | | | 5 | | | | |
| N11 | WE | | | 36 | | | | |
| N12 | $\overline{WE}$ | Frequency | 16 | 11 | 11.49885 | 11.50115 | MHz | 19 |





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NOTE: Load the output under test with the load as specified on figure 13.

FIGURE 19. Frequency test circuit for table XV.



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TABLE XVI. Transition time and pulse width test.

| Test condition | Parameter | | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | Min | Max | | |
| P1 | OUT 1B | Transition time ($t_{THL}$), $C_L$ = 50 pF | Open | 19 | 2.0 | 6.0 | ns | 20,21 |
| P2 | OUT 2B | | | 14 | | | | |
| P3 | OUT 3B | | | 9 | | | | |
| P4 | OUT 4B | | | 5 | | | | |
| P5 | WE | | | 36 | | | | |
| P6 | $\overline{WE}$ | Transition time ($t_{THL}$), $C_L$ = 50 pF | | 11 | | 6.0 | | |
| P7 | OUT 1B | Transition time ($t_{THL}$), $C_L$ = 150 pF | | 19 | | 11.0 | | |
| P8 | OUT 2B | | | 14 | | | | |
| P9 | OUT 3B | | | 9 | | | | |
| P10 | OUT 4B | | | 5 | | | | |
| P11 | WE | | | 36 | | | | |
| P12 | $\overline{WE}$ | Transition time ($t_{THL}$), $C_L$ = 150 pF | | 11 | | 11.0 | | |
| Q1 | OUT 1A | Transition time ($t_{TLH}$), $C_L$ = 50 pF | | 18 | | 15.0 | | |
| Q2 | OUT 2A | Transition time ($t_{TLH}$), $C_L$ = 50 pF | | 13 | | 15.0 | | |
| Q3 | OUT 3A | Transition time ($t_{TLH}$), $C_L$ = 50 pF | Open | 8 | 2.0 | 15.0 | ns | 20,21 |





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TABLE XVI. Transition time and pulse width test - Continued.

| Test condition | Parameter | | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | Min | Max | | |
| Q4 | OUT 4A | Transition time ($t_{TLH}$), $C_L$ = 50 pF | Open | 4 | 2.0 | 15.0 | ns | 20,21 |
| Q5 | WE | Transition time ($t_{TLH}$), $C_L$ = 50 pF | | 36 | | 15.0 | | |
| Q6 | $\overline{WE}$ | Transition time ($t_{TLH}$), $C_L$ = 50 pF | | 11 | | 15.0 | | |
| Q7 | OUT 1A | Transition time ($t_{TLH}$), $C_L$ = 150 pF | | 18 | | 27.0 | | |
| Q8 | OUT 2A | | | 13 | | | | |
| Q9 | OUT 3A | | | 8 | | | | |
| Q10 | OUT 4A | | | 4 | | | | |
| Q11 | WE | | | 36 | | | | |
| Q12 | $\overline{WE}$ | Transition time ($t_{TLH}$), $C_L$ = 150 pF | | 11 | 2.0 | 27.0 | | |
| R1 | OUT 1A | Pulse width ($t_{PW}$) | | 18 | 40.0 | 49.5 | | |
| R2 | OUT 2A | | | 13 | | | | |
| R3 | OUT 4A | | | 8 | | | | |
| R4 | OUT 4A | | | 4 | 40.0 | 49.5 | | |
| R5 | WE | | | 36 | 80.0 | 94.0 | | |
| R6 | $\overline{WE}$ | | Open | 11 | 80.0 | 94.0 | | 20,21 |
| R7 | OUT 1B | Pulse width ($t_{PW}$) | 16 | 19 | 40.0 | 49.5 | ns | 20,22 |



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TABLE XVI. Transition time and pulse width test - Continued.

| Test condition | Parameter | | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | Min | Max | | |
| R8 | OUT 2B | Pulse width ($t_{PW}$) | 16 | 14 | 40.0 | 49.5 | ns | 20,22 |
| R9 | OUT 3B | | | 9 | | | | |
| R10 | OUT 4B | | | 5 | | | | |
| R11 | WE | | | 36 | | | | |
| R12 | $\overline{WE}$ | Pulse width ($t_{PW}$) | 16 | 11 | 40.0 | 49.5 | ns | 20,22 |





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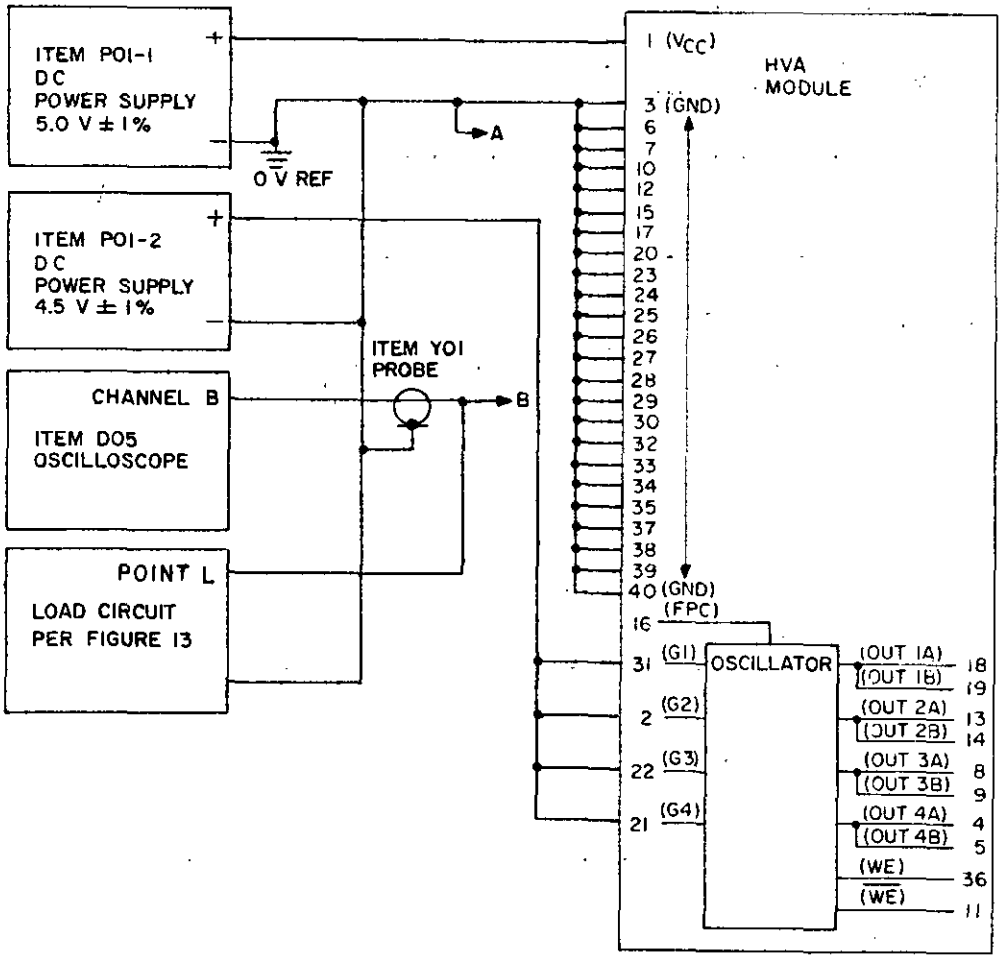



FIGURE 20. Transition time and pulse width test circuit for table XVI.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:17Z
Check the source to verify that this is the current version before use.

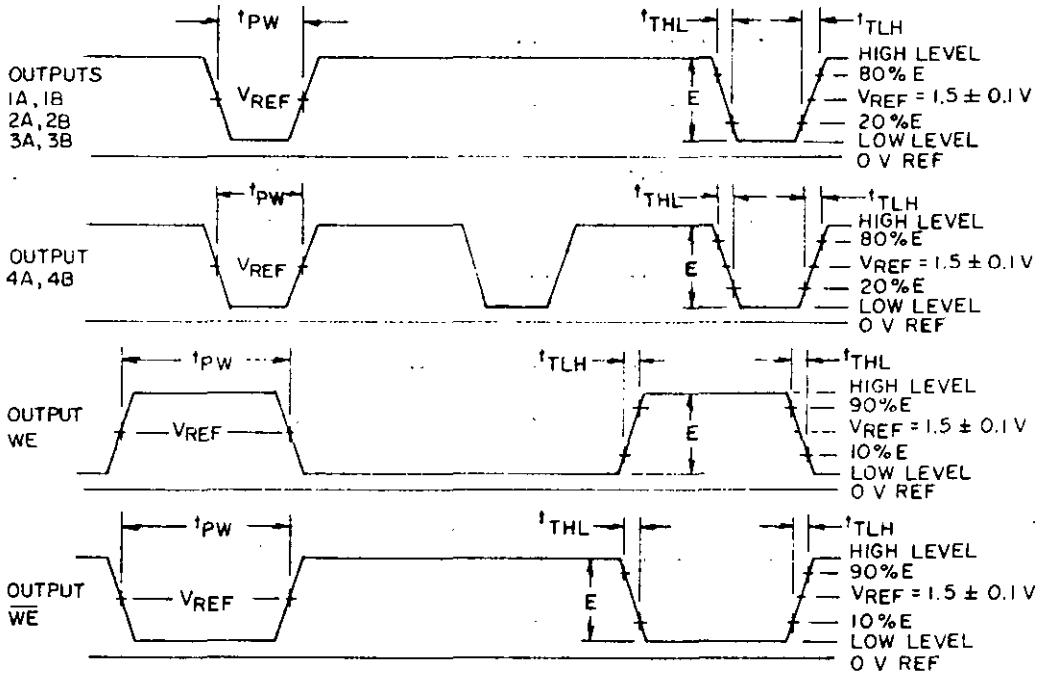



FIGURE 21. Transition time and pulse width test circuit waveforms for table XVI.

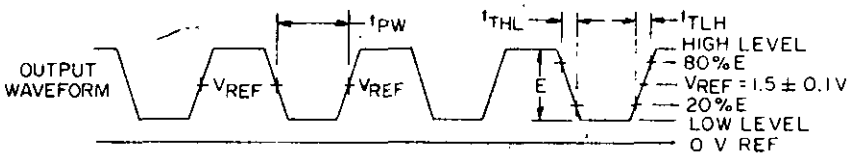



FIGURE 22. Pulse width test circuit waveform for table XVI.



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TABLE XVII. Output to output delay time test.

| Test condition | Parameter | | Connect | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | Min | Max | | |
| S1 | OUT 1A to OUT 2A | Delay time ($t_{DELAY1-LH}$) | Open | 18 | 13 | -2.0 | 2.0 | ns | 23,24 |
| S2 | OUT 2A to OUT 3A | Delay time ($t_{DELAY1-LH}$) | | 13 | 8 | | | | |
| S3 | OUT 1A to OUT 3A | Delay time ($t_{DELAY1-LH}$) | | 18 | 8 | | | | |
| T1 | OUT 1A to OUT 2A | Delay time ($t_{DELAY1-HL}$) | | 18 | 13 | | | | |
| T2 | OUT 2A to OUT 3A | Delay time ($t_{DELAY1-HL}$) | | 13 | 8 | | | | |
| T3 | OUT 1A to OUT 3A | Delay time ($t_{DELAY1-HL}$) | | 18 | 8 | | | | 23,24 |
| U1 | OUT 1A to OUT 4A | Delay time ($t_{DELAY2-LH}$) | | 18 | 4 | | | | 23,25 |
| U2 | OUT 2A to OUT 4A | Delay time ($t_{DELAY2-LH}$) | | 13 | | | | | |
| U3 | OUT 3A to OUT 4A | Delay time ($t_{DELAY2-LH}$) | | 8 | | | | | |
| V1 | OUT 1A to OUT 4A | Delay time ($t_{DELAY2-HL}$) | | 18 | | | | | |
| V2 | OUT 2A to OUT 4A | Delay time ($t_{DELAY2-HL}$) | Open | 13 | 4 | -2.0 | 2.0 | ns | 23,25 |



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TABLE XVII. Output to output delay time test - Continued.

| Test condition | Parameter | | Connect | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | Min | Max | | |
| V3 | OUT 3A to OUT 4A | Delay time ($t_{DELAY2-HL}$) | Open | 8 | 4 | -2.0 | 2.0 | ns | 23,25 |
| W1 | OUT 1A to WE | Delay time ($t_{DELAY3}$) | | 18 | 36 | -14.0 | 2.0 | | 23,26 |
| W2 | OUT 2A to WE | Delay time ($t_{DELAY3}$) | | 13 | 36 | -14.0 | 2.0 | | 23,26 |
| W3 | OUT 3A to WE | Delay time ($t_{DELAY3}$) | | 8 | 36 | -14.0 | 2.0 | | 23,26 |
| X | WE to $\overline{WE}$ | Delay time ($t_{DELAY4}$) | | 36 | 11 | -4.0 | 4.0 | | 23,27 |
| Y | WE to $\overline{WE}$ | Delay time ($t_{DELAY5}$) | Open | 36 | 11 | -3.5 | 2.5 | | 23,27 |
| Z1 | OUT 1A to OUT 2A | Delay time ($t_{DELAY6-LH}$) | 16 | 18 | 13 | -3.0 | 3.0 | | 23,28 |
| Z2 | OUT 2A to OUT 3A | Delay time ($t_{DELAY6-LH}$) | | 13 | 8 | | | | |
| Z3 | OUT 2A to OUT 4A | Delay time ($t_{DELAY6-LH}$) | | 13 | 4 | | | | |
| Z4 | OUT 3A to OUT 4A | Delay time ($t_{DELAY6-LH}$) | | 8 | 4 | | | | |
| Z5 | OUT 1A to OUT 3A | Delay time ($t_{DELAY6-LH}$) | 16 | 18 | 8 | -3.0 | 3.0 | ns | 23,28 |





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TABLE XVII. Output to output delay time test - Continued.

| Test condition | Parameter | | Connect | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | Min | Max | | |
| Z6 | OUT 1A to OUT 4A | Delay time ($t_{DELAY6-LH}$) | 16 | 18 | 4 | -3.0 | 3.0 | ns | 23,28 |
| AA1 | OUT 1A to OUT 2A | Delay time ($t_{DELAY6-HL}$) | | 18 | 13 | | | | |
| AA2 | OUT 2A to OUT 3A | Delay time ($t_{DELAY6-HL}$) | | 13 | 8 | | | | |
| AA3 | OUT 2A to OUT 4A | Delay time ($t_{DELAY6-HL}$) | | 13 | 4 | | | | |
| AA4 | OUT 3A to OUT 4A | Delay time ($t_{DELAY6-HL}$) | | 8 | 4 | | | | |
| AA5 | OUT 1A to OUT 3A | Delay time ($t_{DELAY6-HL}$) | | 18 | 8 | | | | |
| AA6 | OUT 1A to OUT 4A | Delay time ($t_{DELAY6-HL}$) | 16 | 18 | 4 | -3.0 | 3.0 | ns | 23,28 |



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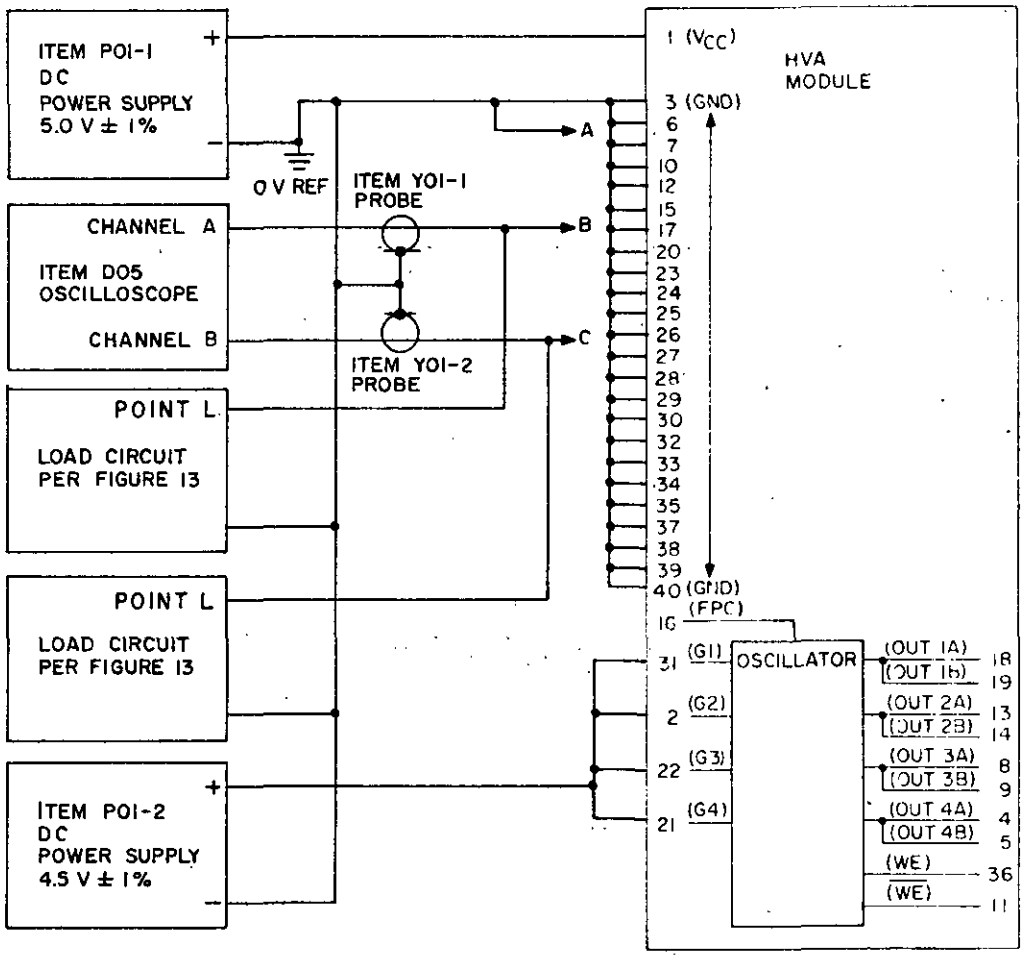



FIGURE 23. Output to output delay time test circuit for table XVII.



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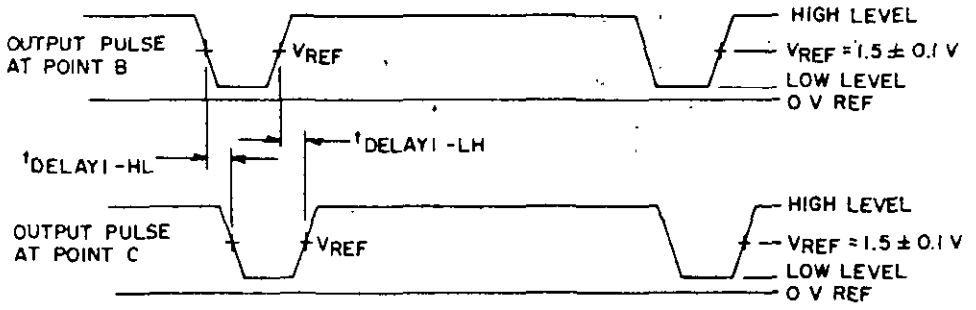



FIGURE 24. Output to output delay time test circuit waveforms for table XVII, test conditions S1-S3 and T1-T3.

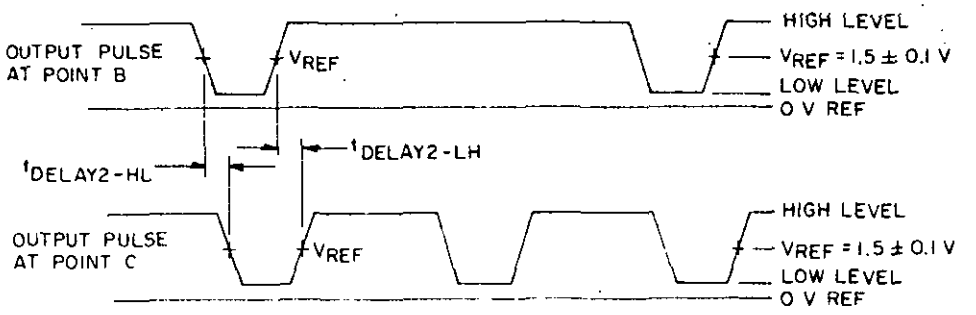



FIGURE 25. Output to output delay time test circuit waveforms for table XVII, test conditions U1-U3 and V1-V3.



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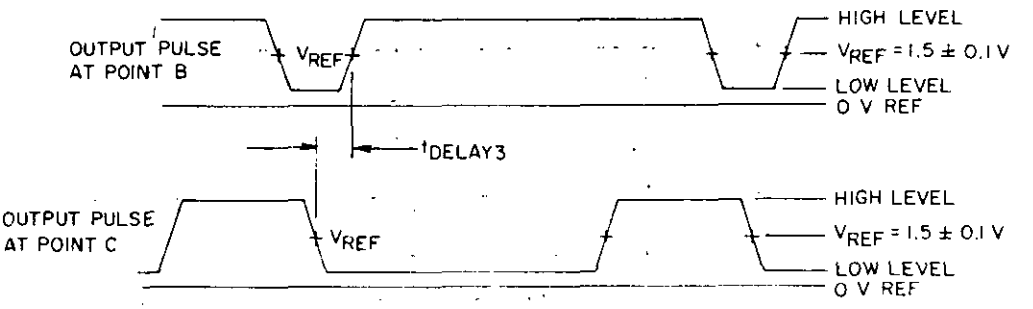



FIGURE 26. Output to output delay time test circuit waveforms for table XVII, test conditions W1-W3.

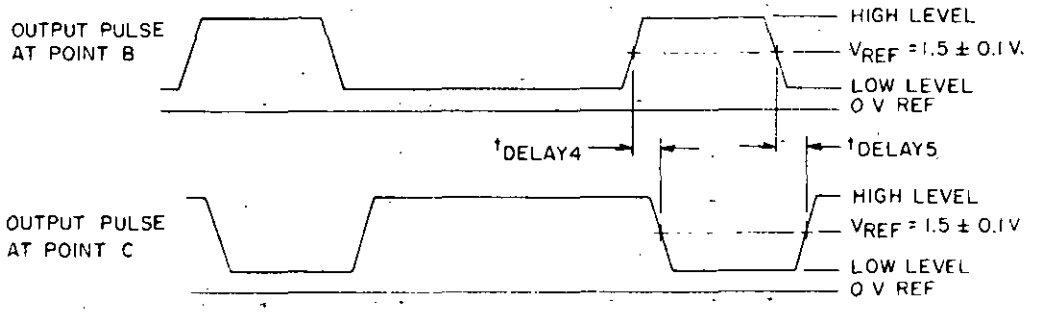



FIGURE 27. Output to output delay time test circuit waveforms for table XVII, test conditions X and Y.

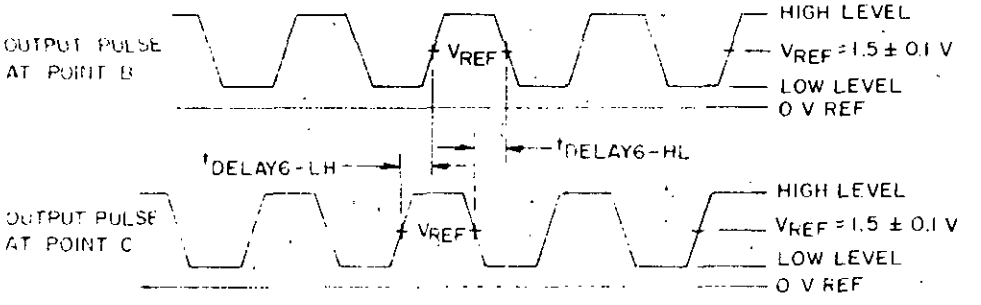



FIGURE 28. Output to output delay time test circuit waveforms for table XVII, test conditions Z1-Z6 and AA1-AA6.



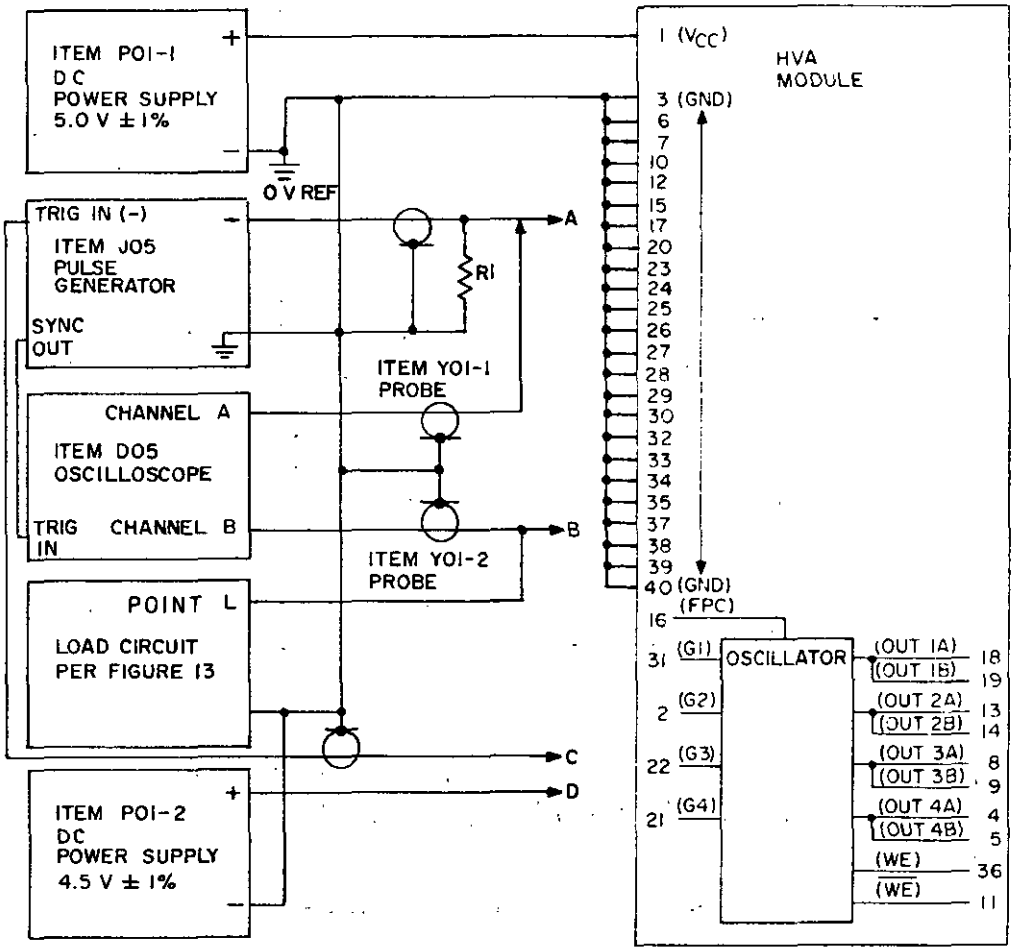
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Check the source to verify that this is the currentversion before use.

TABLE XVIII. Propagation delay time (gate to output) test.

| Test condition | Parameter | | Connect | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| BB1 | G1 to OUT 1A | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | 31 | 18 | 8 | 2,22, 21 | 2.0 | 8.5 | ns | 29,30 |
| BB2 | G2 to OUT 2A | | 2 | 13 | 8 | 31,22, 21 | | | | 29,30 |
| BB3 | G3 to OUT 3A | | 22 | 8 | 18 | 31,2, 21 | | | | 29,30 |
| BB4 | G4 to OUT 4A | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | 21 | 4 | 18 | 31,2, 22 | 2.0 | 8.5 | | 29,31 |
| BB5 | G1 to OUT 1A | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | 31 | 18 | 8 | 2,22, 21 | 4.0 | 12.5 | | 29,30 |
| BB6 | G2 to OUT 2A | | 2 | 13 | 8 | 31,22, 21 | | | | 29,30 |
| BB7 | G3 to OUT 3A | | 22 | 8 | 18 | 31,2, 21 | | | | 29,30 |
| BB8 | G4 to OUT 4A | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | 21 | 4 | 18 | 31,2, 22 | 4.0 | 12.5 | ns | 29,31 |



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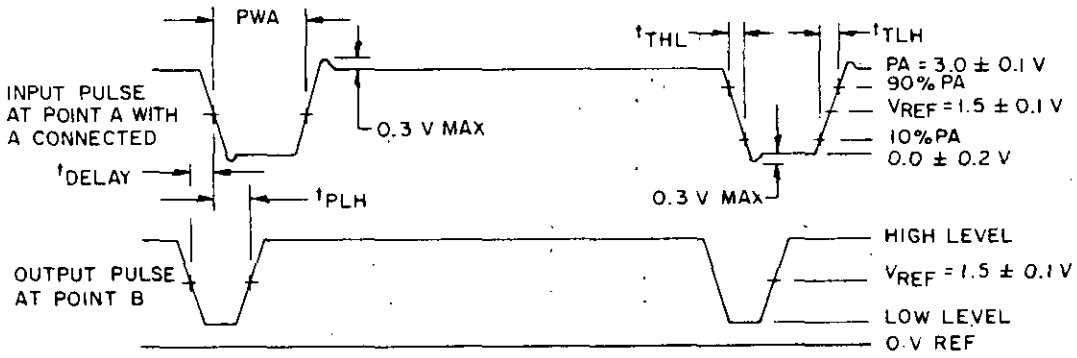



NOTE: R1 = 51.1Ω ±1%, 1/4 W.

FIGURE 29. Propagation delay time (gate to output) test circuit for table XVIII.



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Check the source to verify that this is the current version before use.

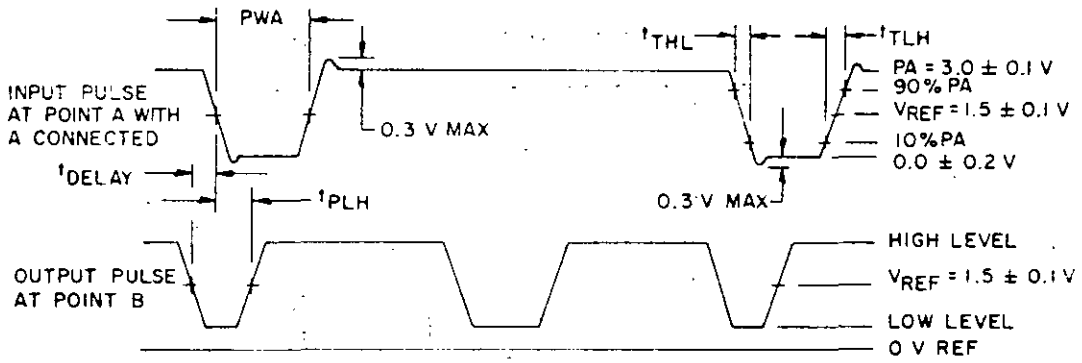



NOTES:
1. Input pulse parameter:
   The input pulse is taken from the negative output of item J05. Adjust the dc offset and pulse amplitude to obtain the required pulse voltage levels. Adjust the delay control of item J05 to obtain:
   $t_{DELAY}$ = 20 ±5 ns.
   PWA = 50 ±5 ns.
   $t_{TLH} = t_{THL}$ = 4 ±2 ns.
2. Verify that all outputs have a high level $\geq$ 2.0 V and a low level $\leq$ 0.5 V before making measurements.
3. The physical length of conductors between resistor R1 and input pin; between items Y01-1 and Y01-2 and input pins; and between load circuit and output pin, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 30. Propagation delay time (gate to output) test circuit waveforms for table XVIII.



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Check the source to verify that this is the current version before use.




NOTES:
1. Input pulse parameters:
   The input pulse is taken from the negative output of item J05. Adjust the dc offset and pulse amplitude to obtain the required pulse voltage levels. Adjust the delay control of item J05 to obtain:
   $t_{DELAY}$ = 20 ±5 ns.
   PWA = 50 ±5 ns.
   $t_{TLH} = t_{THL}$ = 4 ±2 ns.
2. Verify that all outputs have a high level $\geq$ 2.0 V and a low level $\leq$ 0.5 V before making measurements.
3. The physical length of conductors between resistor R1 and input pin; between items Y01-1 and Y01-2 and input pins; and between load circuit and output pin, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 31. Propagation delay time (gate to output) test circuit waveforms for table XVIII.



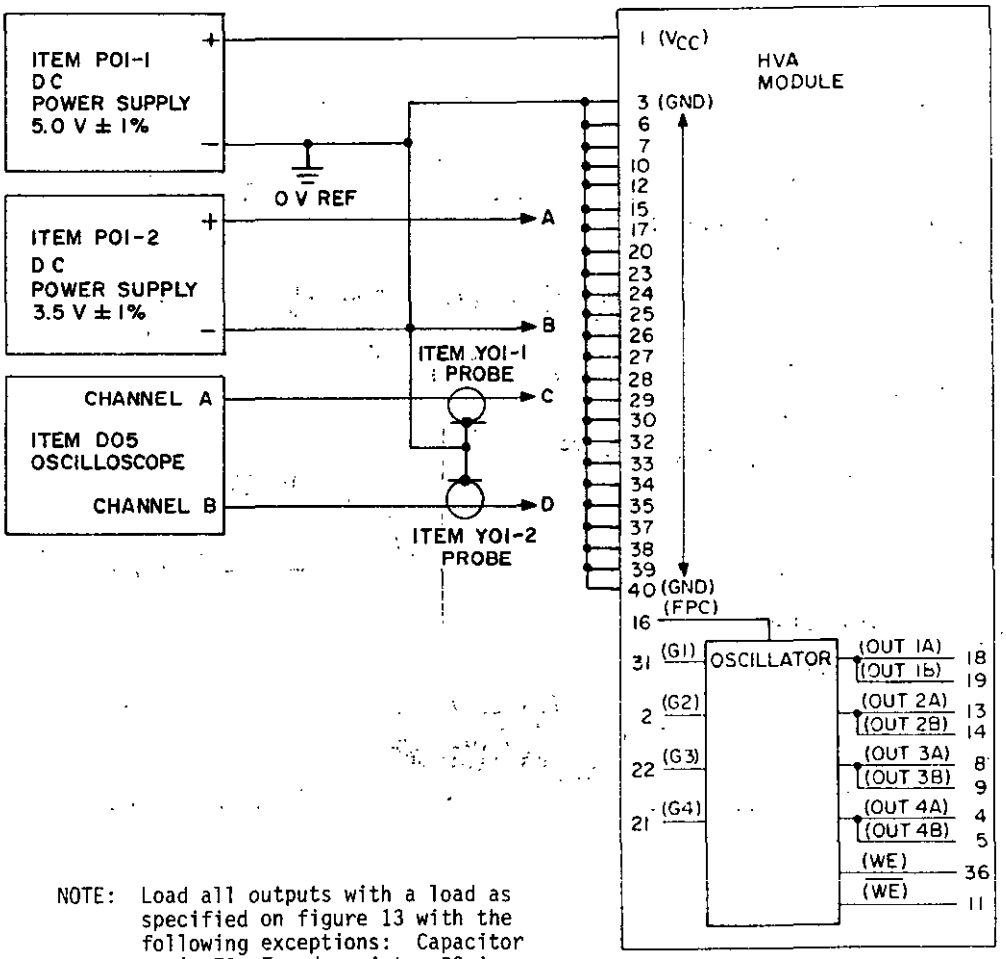
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Check the source to verify that this is the current version before use.

TABLE XIX. Output crosstalk (high-level and low-level) test.

| Test condition | Parameter | | Connect | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| CC1 | OUT 1A, 1B | Output crosstalk, high level | 2,21, 22 | 31 | 18,19 | 8 | 0 | 400 | mV | 32,33 |
| CC2 | OUT 2A, 2B | | 21,22, 31 | 2 | 13,14 | 8 | | | | |
| CC3 | OUT 3A, 3B | | 2,21, 31 | 22 | 8,9 | 13 | | | | |
| CC4 | OUT 4A, 4B | | 2,22, 31 | 21 | 4,5 | 8 | | | | |
| CC5 | WE | | 2,21, 22,31 | Open | 36 | | | | | |
| CC6 | $\overline{WE}$ | Output crosstalk, high level | | | 11 | | | 400 | mV | 32,33 |
| DD1 | OUT 1A, 1B | Output crosstalk, low level | | | 18,19 | | | 1.0 | V | 32,34 |
| DD2 | OUT 2A, 2B | | | | 13,14 | 8 | | | | |
| DD3 | OUT 3A, 3B | | | | 8,9 | 18 | | | | |
| DD4 | OUT 4A, 4B | | | | 4,5 | | | | | |
| DD5 | WE | | | | 36 | | | | | |
| DD6 | $\overline{WE}$ | Output crosstalk, low level | 2,21, 22,31 | Open | 11 | 18 | 0 | 1.0 | V | 32,34 |



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NOTE: Load all outputs with a load as specified on figure 13 with the following exceptions: Capacitor $C_L$ is 50 pF and resistor R2 is 825Ω for outputs 1A-4A.

FIGURE 32. Output crosstalk (high-level and low-level) test circuit for table XIX.



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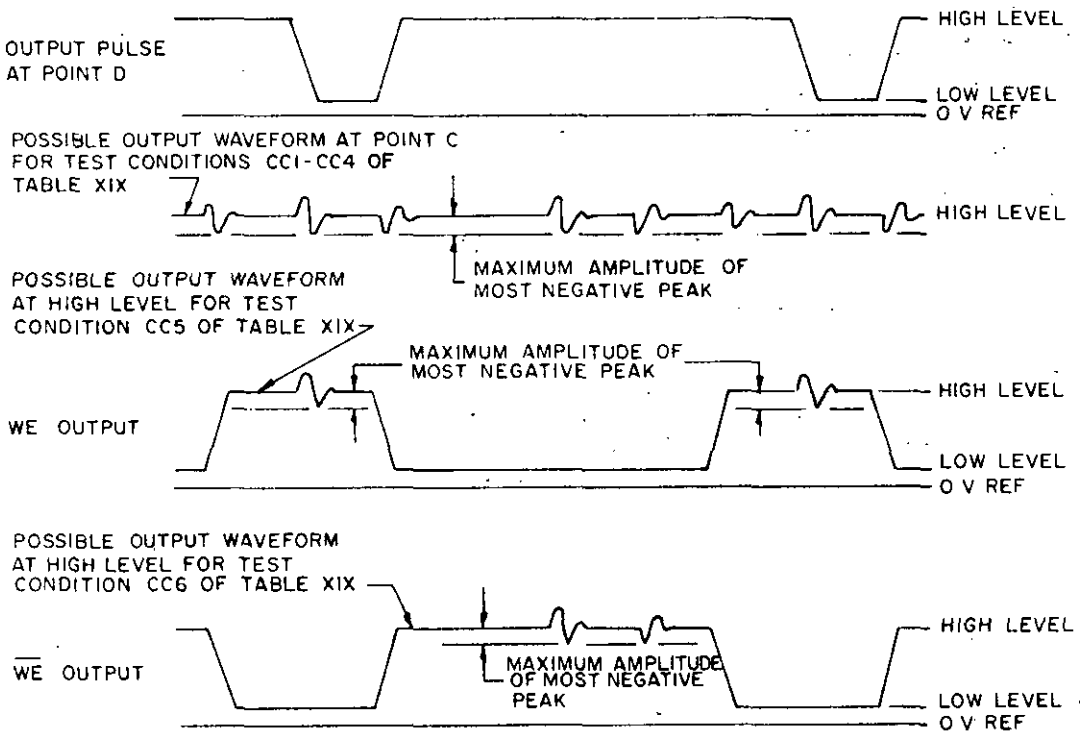



FIGURE 33. Output crosstalk (high-level) test circuit waveforms for table XIX.



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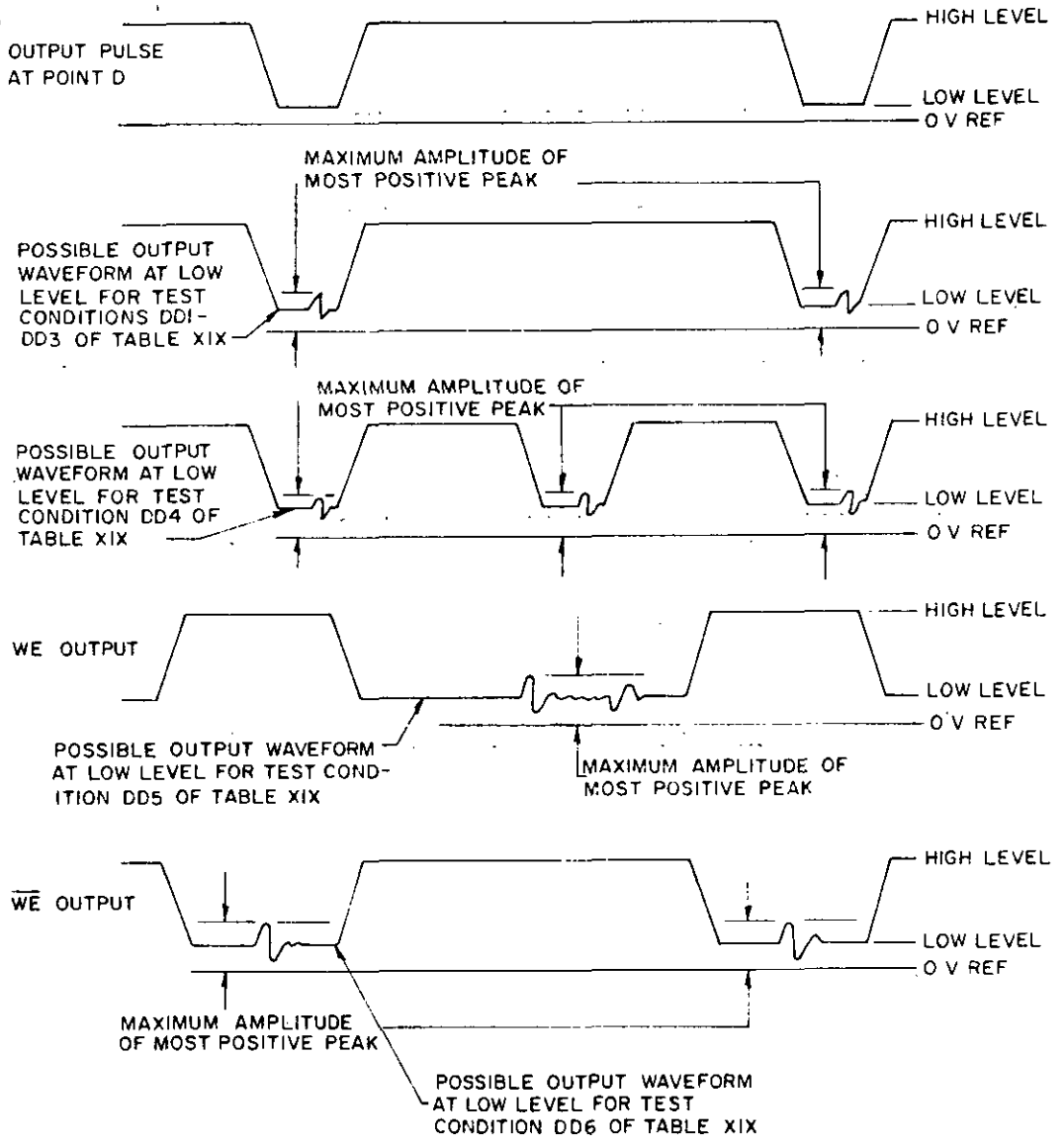



FIGURE 34. Output crosstalk (low-level) test circuit waveforms for table XIX.



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TABLE XX. Power supply transient amplitude test.

| Test condition | Parameter | Connect | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | A to pin | Min | Max | | |
| EE | Power supply transient amplitude | 18 | 0 | 500 | mV | 35,36 |



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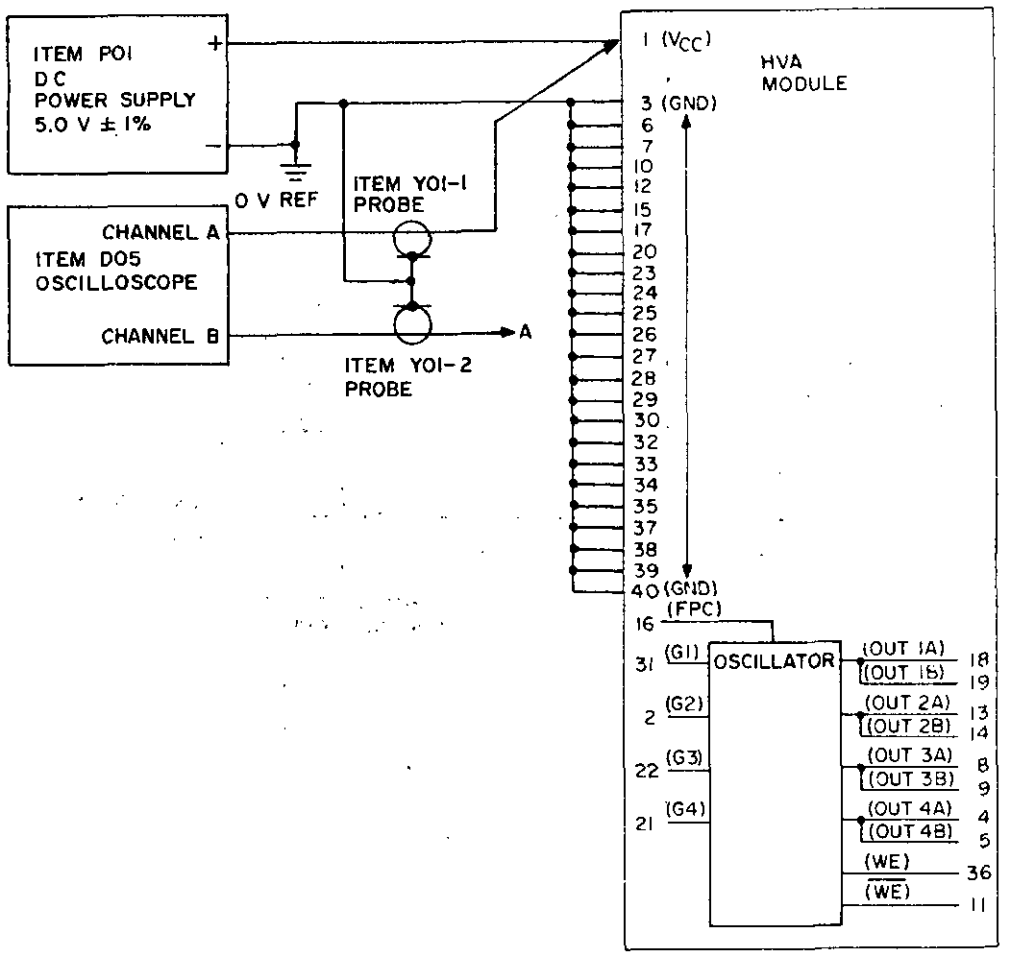



FIGURE 35. Power supply transient amplitude test circuit for table XX.



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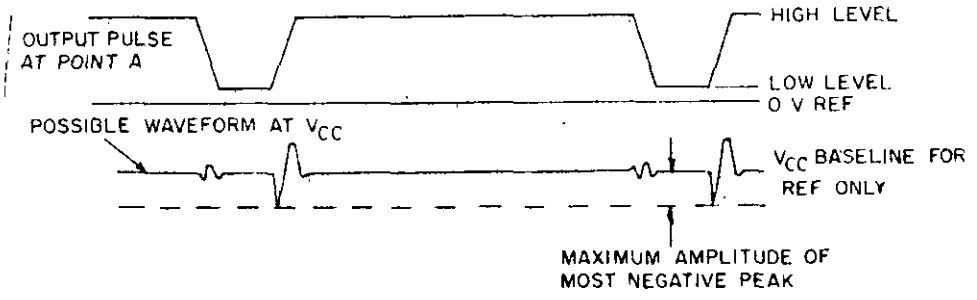



FIGURE 36. Power supply transient amplitude test circuit waveforms for table XX.



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TABLE XXI. Power supply filter capacitance test.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| FF | Power supply filter capacitance | Open | 1 | 4.4 | 11.6 | µF | 37 |





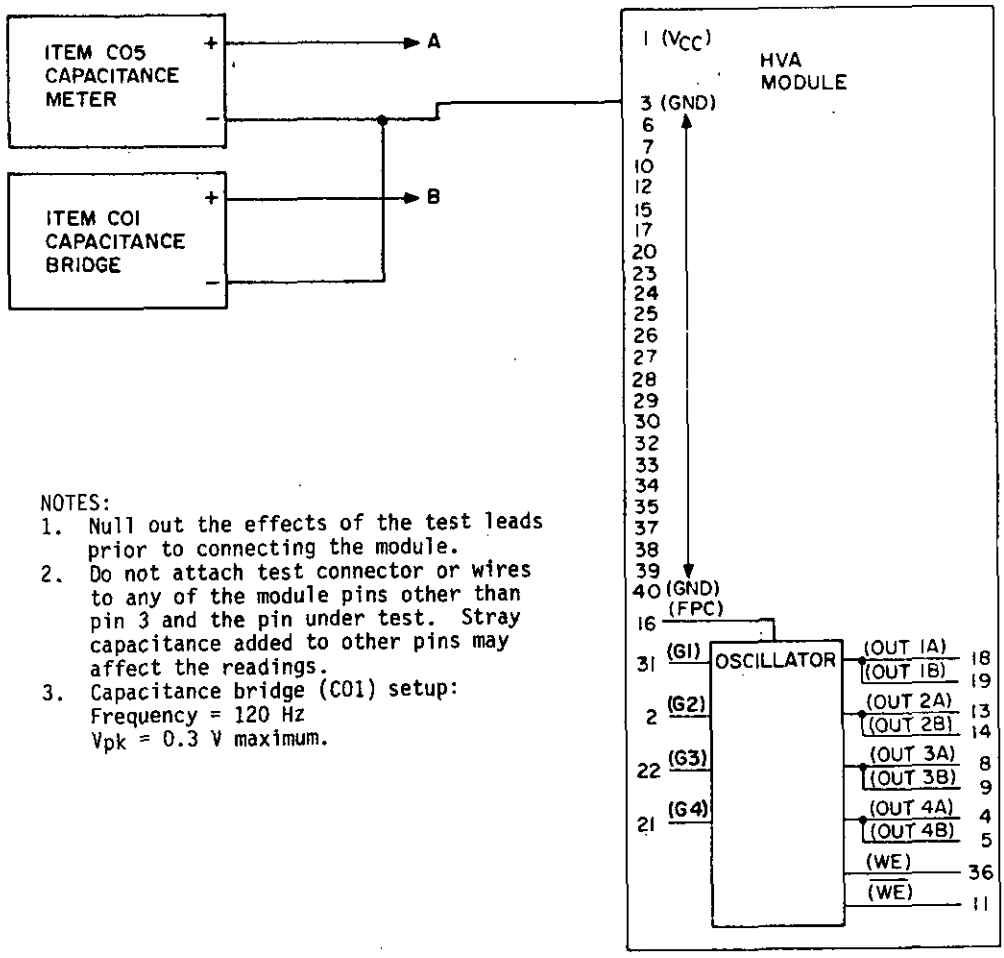
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TABLE XXII. Input capacitance test.

| Test condition | Parameter | | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | Min | Max | | |
| GG1 | G1 | Input capacitance | 31 | Open | 5.0 | 20.0 | pF | 37 |
| GG2 | G2 | ↓ | 2 | ↓ | ↓ | ↓ | ↓ | ↓ |
| GG3 | G3 | ↓ | 22 | ↓ | ↓ | ↓ | ↓ | ↓ |
| GG4 | G4 | Input capacitance | 21 | Open | 5.0 | 20.0 | pF | 37 |



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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 3 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (CO1) setup:
   Frequency = 120 Hz
   $V_{pk}$ = 0.3 V maximum.

FIGURE 37. Power supply filter and input capacitance test circuit for tables XXI and XXII.



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